EVIDENCESYNC: INTELLIGENT CCTV EVIDENCE ANALYSIS AND MULTI-CAMERA INCIDENT RECONSTRUCTION SYSTEM

PAMCA698J - Dissertation-1

Submitted in partial fulfillment of the requirements for the degree of

Master of Computer Applications

in

Department of Computer Applications

by

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25MCA0248

Under the guidance of

Dr. Jagadeesan S

School of Computer Science Engineering and Information Systems

VIT, Vellore



November, 2026

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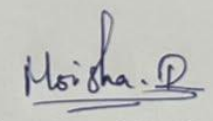
November, 2026

DECLARATION

I hereby declare that the **PAMCA698J - Dissertation-1** entitled **"EvidenceSync: Intelligent CCTV Evidence Analysis and Multi-Camera Incident Reconstruction System"** submitted by me, for the award of the degree of Master of Computer Applications in Department of computer Application, School of Computer Science Engineering and Information Systems to VIT is a record of bonafide work carried out by me under the supervision of **Dr. Jagadeesan S**, Associate Professor Sr. Grade 1, SCORE, VIT, Vellore.

I further declare that the work reported in this dissertation has not been submitted and will not be submitted, either in part or in full, for the award of any other degree ordiploma in this institute or any other institute or university.

Place: Vellore
Date:



Signature of the Candidate

CERTIFICATE

This is to certify that the **PAMCA698J - Dissertation-1** entitled **"EvidenceSync: Intelligent CCTV Evidence Analysis and Multi-Camera Incident Reconstruction System"** submitted by **Monisha I (25MCA0248)**, SCORE, VIT, for the award of the degree of Master of Computer Applications in Department of Computer Applications, is a record of bona fide work carried out by him under my supervision during the period, 15. 07. 2026 to 28.10.2026, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute orany other institute or university. The dissertation fulfills the requirements and regulations ofthe University and in my opinion meets the necessary standards for submission.

Place: Vellore
Date:

23/10/2026

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Executive Summary

The project focuses on intelligent CCTV evidence analysis and multi-camera incident reconstruction. During implementation, the core system was developed using computer vision, deep learning, object detection, object tracking, and video analysis techniques. Designed to perform analyses of CCTV videos recorded with four cameras.For this solution,YOLO11n was used to detect objects in videos,and ByteTrack was applied to track detected objects.Thus,the reults of objects detections,tracking,and timeline events were stored for further inspection.FastAPI backend was implemented to provide access to analysis results,and a web dashboard was created to videw ideos in browser,inspect detection results ,and visualize two-deimensional movement of tracked objects.By consolidating this information in one place,EvidenceSync facilitates investigators in reviewing recorded events captured with multiple CCTV cameras.With the current set of features,EvidenceSync allows focusing on reviewing CCTV footage recorded from multiple angles.In the future,it could be interesting to add suspicious activity detection,human pose estimation,extract relevant parts of videos for each camera,and create a synchronized version of a video with events.This would allow streamlining CCTV footage review and reducing the amount of manual work compared to the current solutions.

Keywords: Artificial Intelligence, Computer Vision, Deep Learning, YOLO, ByteTrack, CCTV Surveillance, Multi-Camera Analysis, Human Pose Estimation, Incident Reconstruction.

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CCTV	Closed-Circuit Television
CSS	Cascading Style Sheets
CPU	Central Processing Unit
FPS	Frames Per Second
HTML	HyperText Markup Language
HTTP	Hypertext Transfer Protocol
IoU	Intersection over Union
JSON	JavaScript Object Notation
OpenCV	Open Source Computer Vision Library
RAM	Random Access Memory
YOLO	You Only Look Once
YOLO11n	YOLO11 nano model variant
2D	Two-dimensional

Symbols and Notations

t	Timestamp or elapsed time of an observation
x,y	Two-dimensional image/trajectory coordinates
w,h	Bounding-box width and height
c	Detection confidence score
(x1,y1,x2,y2)	Bounding-box corner coordinates
f	Video frame index
Δt	Elapsed time between observations
ID	Camera-local tracking idetifier

CHAPTER 1

INTRODUCTION

Video surveillance is an essential part of any security and incident investigation system in public places, organizations, and other objects. However, a significant amount of video data should be reviewed to find relevant activities, moving objects, and investigate what exactly happens. For this purpose, there is a need for an intelligent system that can understand and analyze video evidence.

A standard closed-circuit television system records videos that are then reviewed by investigators. For multiple CCTV cameras, the analysis is more complicated because it is necessary to consider all video streams and correlate events that occur in them. Computer vision and deep learning methods can help in automated CCTV evidence analysis by recognizing and tracking objects in videos. It can support an investigator by providing activity details in a specific area captured by a camera. In addition, such a system should create a timeline of activities to help review what happened during the incident.This project aims to design EvidenceSync: Intelligent CCTV Evidence Analysis and Multi-Camera Incident Reconstruction System. It will use computer vision to enable automated CCTV evidence analysis for multiple cameras. The object detection model based on the You Only Look Once (YOLO) architecture was trained to detect objects in video evidence. ByteTrack was used to track detected objects in video frames and provide details about their movement. The system has a web dashboard that allows investigators to review and analyze video evidence and track detected objects using FastAPI web framework. Designed functionality includes a multi-camera viewer, object detection visualization, timeline review, and 2D trajectory analysis.

The current implementation contains a prototype of a system that demonstrates object detection and tracking for four CCTV cameras. Detected objects are visualized with bounding boxes, and the dashboard displays detection and tracking results, including a timeline of events. The next step is to implement functionality for finding suspicious activities or objects. It should also be possible to select particular time intervals from different cameras and analyze them in the context of the entire event. There is a need for posture recognition to estimate how people move in space captured by multiple cameras. Finally, the system should create a synchronized incident replay video out of multiple cameras.

1.1 Objective

The main objective of this project is to create an artificial intelligence system capable of analyzing CCTV evidence. Specifically, the application to be designed should enable object detection, object tracking, and multi-camera reconstruction of events.

The objectives of the project, therefore, include:

- Processing CCTV videos to extract necessary information.
- Integrating a YOLO model to detect objects in the video.
- Using ByteTrack to track the detected objects.
- Storing the tracked objects events for further processing and analysis
- Creating a FastAPI backend to display results of the CCTV analysis.
- Developing a frontend dashboard to enable the simultaneous viewing of multiple videos.
- Displaying outputs of the object detection and two-dimensional movement tracking in a vedio.
- Creating a system dashboard a view objects of interest and reconstructed events in four different camera views.
- Investigating methods to detect suspicious activities and human pose estimation to reconstruct events .
- Developing a method to extract relevant video segments from different cameras and combine them into a single video for easier viewing and anlysis.
- Testing the developed methods and analyzing the results

The objectives can be grouped into several categories. The first category includes the goals of developing a multi-camera CCTV video analysis system. The second category comprises the goals of utilizing the object detection and tracking capabilities of AI to identify objects of interest, track them, and display the results. Similarly, the third category includes the objectives of using time-event processing to identify objects of interest and analyze their movements. The last category is the goal of using the application for the intended purpose. In summary, the objectives of this project were to develop an AI-based CCTV video analysis application. The developed system should be able to track objects, analyze events, and reconstruct activities

using multiple camera views. The application is currently only able to view and analyze recorded evidence and display the results of the tracking and event analysis. Investigating and developing advanced methods of object detection, analyzing suspicious activities, and reconstructing events using AI is recommended for further research. The system could also be improved by enabling it to automatically combine multiple videos into a single video. The current work, however, does not claim to have fully tested and developed an application that can perform 3-D reconstruction of events from CCTV videos..

1.2 Motivation

As CCTV surveillance becomes increasingly prevalent in society, so does the need to efficiently analyze and investigate the recorded footage. While conventional CCTV surveillance systems merely serve the purpose of recording, investigation often requires reviewing and correlating video data from multiple cameras, which is a resource-intensive and time-consuming task.On the one hand, existing object detection approaches allow one to detect objects in a given frame. However, the detection of objects is only a part of the analysis needed, as investigators often need to understand how the objects move and what events occur in the video. Furthermore, investigators may want to analyze several video streams simultaneously or review several clips; in this case, manually correlating the events in different video streams is difficult and requires much time to perform.

With the existing algorithms and models from the fields of computer vision and deep learning, it is possible to attempt to design and implement an efficient algorithm for reviewing and investigating CCTV footage. Thus, one can use object detection models to (re)detect objects in the video frames and object-tracking algorithms to track the objects in video frames to identify which objects move in the field of view and where they move. Similarly, one can use timeline analysis and synchronized video playback to correlate the events. However, while tracking algorithms can track objects within one camera view, they do not provide a guarantee that the object in view is the same as the one that was previously tracked in another camera view.This limitation outlines the motivation for this project, which is to design and implement an algorithm that allows investigators to efficiently review and correlate multiple CCTV video streams. EvidenceSync is a conceptual design for an intelligent CCTV evidence analysis and incident reconstruction system. It is built with the core components of the You Only Look Once (YOLO) object detection model and the ByteTrack object-tracking model, a FastAPI backend

API, and a frontend dashboard to enable investigators to view and analyze the CCTV video evidence, view detected objects, analyze and view timeline data, and view two-dimensional object movement trajectories.The motivation for this project is to minimize the investigators' workload related to manually reviewing CCTV video footage. Future work to expand upon this project could involve detecting and analyzing potentially suspicious activities, determining body pose data for people, determining time intervals of interest with respect to all camera views, and creating and viewing synchronized video of an incident from multiple camera views.

1.3 Background

1.3.1 Object Detection in CCTV Surveillance

Object detection is a computer vision method responsible for detecting and recognizing objects in a video frame. In the case of CCTV surveillance, it allows for detecting objects of interest and tracking them without requiring a human operator to watch hours of video. Traditionally, the process of analyzing a video recording relied on the manual inspection conducted by human operators. With modern surveillance involving numerous cameras and lengthy video recordings, such a method becomes inefficient.Automatic and semi-automatic methods of video analysis rely on the principles of deep learning and convolutional neural networks. These methods are based on the architecture of the You Only Look Once, or YOLO, family of neural networks. They allow for detecting and classifying objects by creating bounding boxes around the objects of interest in a given frame. Thus, such networks enable automatic and semi-automatic video analysis, which is beneficial for CCTV surveillance and security.

It is important to note that any object detection model benefits from an optimal set of parameters, which include the quality and format of the input video, lighting conditions, and object size and orientation. Therefore, a video frame with high quality, optimal lighting, and clear depictions of objects of interest is more likely to produce accurate results. The main benefit of using such AI models is that they do not raise a false alarm, as a single detected object does not imply any suspicious activity. The two should be taken in context with the object's behavior and the events taking place in the video.

In the scope of this project, EvidenceSync uses an AI-based object detection system based on the YOLO architecture. It allows for detecting objects of interest and storing this information as a set of bounding box annotations. As a result, the object detection system provides the necessary input for further processing and analysis, which includes object tracking, timeline creation, and 2D object reconstruction within the multi-cam reconstruction dashboard.

1.3.2 Object Tracking Using Deep Learning

In computer vision, object tracking refers to the set of algorithms that allow the association of the same object detected in two successive video frames (from a single camera or multiple cameras). While object detection deals with identifying objects in single images, object tracking adds a temporal element to detection: an object detection algorithm can detect all the objects present and provide their coordinates and identities in each frame. However, the results of the detection algorithm for successive frames need to be associated to infer information about the movement of objects, such as speed and direction. This information is critical in video surveillance tasks such as CCTV monitoring and plays an important role in reconstructing events from the video records.

ByteTrack is an algorithm for object tracking that assigns detection results to tracks using motion information and detection confidence. However, there are several challenges when tracks need to be maintained across video frames, such as occlusions, deformations, ambiguities in dense scenes, and missed detections. Furthermore, the tracks obtained by a video surveillance system are typically only valid for a single video source: there is currently no standard way to define the same track identity across different cameras.In EvidenceSync, the ByteTrack object tracking algorithm is used in combination with a YOLO-based object detector to track detected objects across video frames. This information can then be used to display bounding boxes around the objects of interest, analyze their movements, and visualize them using the reconstruction dashboard. It is the basis for exploring the movements of objects of interest within the field of view of a camera.

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1.3.3 Multi-Camera Surveillance Analysis

Multi-camera surveillance analysis consists of examining video recordings captured by multiple cameras in order to gain a better general understanding of the recorded environment. As any single camera only observes a limited perspective, investigators may use multiple cameras to gain additional insight into the recorded environment and review objects or actions that would otherwise be impossible to observe.There are multiple considerations when analyzing video recordings captured by multiple cameras, including individual fields of view, varying levels of detail, differences in recording time, and differences in recording duration. By syncing the playback of videos captured by different cameras, an investigator can directly compare the observed activity in their respective field of view at a specific elapsed time since the beginning of the recording. However, in order to properly sync videos captured by different cameras, the beginning of the recordings has to be defined, and the videos have to be adjusted in accordance with any clock drift between different cameras. Similarly, identifying the same objects or people in different cameras adds an additional layer of complexity to the analysis.As a result, in the current version of EvidenceSync, 4 different views of CCTV cameras have been combined into a single reconstruction dashboard, which also includes synchronized playback controls. The dashboard displays saved object detection and tracking results for each camera view, allowing an investigator to not only review the recorded activities from different perspectives, but also to examine the program's processing results in more detail.

The current version of the dashboard offers multi-camera views with synchronized elapsed time playback controls. Additional features that would be beneficial for multi-camera surveillance analysis include cross-camera identification of individuals and objects and reconstruction of a person's full movement path across cameras, which could be implemented in future versions of the program.

1.3.4 Human Activity Recognition and pose Estimation

Human activity recognition is a computer vision technique that allows you to identify and classify activities of people on video. For CCTV surveillance, it can be useful to analyze video data and identify activities such as walking, running, bending, falling, etc. This information may be helpful during the review of video evidence and investigation of a possible crime.Human pose estimation is a computer vision task that involves identifying key body points such as shoulder, elbow, wrists, hips, knees, and ankles. Using this information, it is possible to track specific body parts and analyze the movement of a person, as a general idea, during each frame of a video. Pose estimation may be helpful for investigations because it can add informative evidence to understand the recorded events.However, it is important to note that such activity recognition should be considered as a part of a bigger picture. Information about the detected activities depends on the quality of the source video, the lighting conditions during the recording, the viewing angle, obstructed parts of the body, and a number of other factors. Additionally, by itself, human activity recognition cannot confirm or deny a person's guilt. It should always be considered in the context of other information and video evidence.

In the context of the EvidenceSync project, this technology can be considered as a possible improvement to the current level of CCTV video analysis. The current feature set focuses on detecting objects, tracking them during the whole video, reviewing events in a timeline view, and analyzing 2D movement patterns. It will be helpful to consider adding such features in the future that allow detecting human body key points and analyzing the activities of the observed persons.

1.3.5 Video-Based Incident Reconstruction

The video-based incident reconstruction technique consists of analyzing recorded surveillance videos with the objective to comprehend what has transpired in the context of a particular incident. It involves the multimedia analysis technology of video content analysis and utilizes object detection, video tracking algorithms, and temporal information to recognize relevant events and analyze their timeline. With multiple camera views, each showing the incident from a different perspective, the ability to navigate between them in a way that reconstructs the temporal development of the incident enables the investigation.

The key to the efficient reconstruction of an incident that has occurred is the manual or automatic search in time-consuming video content for relevant information. Particularly, when multiple cameras are involved, investigators may need to watch several videos to establish the chronology of events. A better organization of the detected objects in time and the ability to navigate through them may help focus on the important parts of the recording and aid the investigative process.The precision of video-based incident reconstruction depends on the performance of object detection and tracking algorithms and the accuracy of timestamps. Moreover, if the cameras are not synchronized, different viewpoints will show the incident differently due to the relative position of objects and humans in space. As such, it is essential to remember that video reconstruction produces estimates needed to determine the course of an event based on available evidence.

In EvidenceSync, the video-based incident reconstruction will be implemented as a web dashboard, enabling an investigator to view four camera views in a synchronized view, navigate through time using the playback controls, see the detected objects and events in a timeline view, and observe the movement of people and objects in a 2D reconstructed environment. The temporal context of the video content will be augmented with automatically extracted events related to the incident under review. Additionally, in future developments, the system may feature automated incident interval extraction, human pose reconstruction, and a multi-angle video composition utility to create a composite video of the reconstructed incident.

1.3.6 Related Work

Research in the field of intelligent video surveillance has focused on object detection, object tracking, human activity recognition, and multi-camera incident analysis. These areas of study tackle various sides of identifying objects, figuring out how they moved, and analyzing video evidence. With that said, combining all of them can effectively support the process of reviewing CCTV footage and investigating incidents.The literature related to this project falls into four categories: object detection in video surveillance, object tracking and motion analysis, multi-camera video surveillance, and human activity recognition and video-based incident reconstruction.

1.3.6.1 Object Detection in CCTV Surveillance

Object detection lies at the heart of automated video analysis. YOLO family model perform object detection tasks by predicting the classes and the locations of the detected objects in a video frame. These models have been applied in a wide range of computer vision applications that require an efficient analysis of video contents. In EvidenceSync, a YOLO-based model is used to detect the objects present in a CCTV video and overlay bounding-boxes on top of them. The detected objects play a pivotal role in further tracking and review of evidence

1.3.6.2 Object Tracking and Movement Analysis

Object tracking aims to associate detections from different frames to estimate their movements. ByteTrack is a tracking algorithm based on detection information that links objects detected in different frames. Tracking can help to visualize movements and analyze recorded activities. In EvidenceSync, it is used together with object detection to create camera-local tracking results and display movement paths. It is important to note, however, that in order to ensure that objects detected in different cameras are the same, it is necessary to match objects detected in different cameras according to their identities.

1.3.6.3 Multi-Camera Surveillance and Incident Analysis

Multi-camera surveillance offers varying perspectives of the same environment which can be leveraged by investigators looking into a case that is presented in multiple angles by individual recordings. This is done by playing the videos in sync and analyzing them together on a timeline, yet the challenge is presented in the potential differences between cameras. EvidenceSync currently allows for the syncing and reviewing of up to 4 cameras on the same dashboard to help synchronize the playback and analyze the saved results of any detections, but the ability to associate identities across different sources is yet to be perfected.

1.3.6.4 Human Activity Recognition and Pose Estimation

Human activity recognition and pose estimation are key research areas in intelligent surveillance. The former focuses on recognising observable activities from video sequences while the latter estimates joints to represent the posture and movement of the human body. It

is possible that such methods would be helpful in providing additional context to incidents under investigation. However, human pose estimation and automated activity classification are features that are proposed to be added in the future of EvidenceSync and not yet implemented.

1.3.6.5 Comparative Analysis of Existing Research

The object detection, tracking, multi-camera visualization, and human activity recognition are different aspects of CCTV investigation. As demonstrated by EvidenceSync, the utilization of object detection, ByteTrack tracking, timeline review, synchronized playback, and 2D movement visualization in a unified system allows for advanced scrutiny of any specific recorded evidence. In its current state, the system provides an effective basis for further consideration of video evidence in 4 camera views.The main strength of the current system is the utilization of all of these aspects within one piece of software designed to review evidence. In the future, it could be further improved by introducing a feature that highlights potentially suspicious activities.In additional, the system could be expanded to include the body posture by tracking of the human and timelines from multiple cameras to synchronize the incident.These attributes would be enable the in-depth examination of activities and offer an enhanced multi-camera incident reconstruction

Table 1.1 Comparative Analysis of Existing Research

S. No.	Title of the Research Paper	Author(s) and Year	Methodology	Key Findings
1	You Only Look Once: Unified,Real-Time Object Dectection	Redmon et al(2016)	UOLO-based object detecion	Introduces real-time object detection using a single neutral network.Relevant to detecting objects in CCTV frames.
2	SSD:Single Shot MultiBox Detector	Liu et al. (2016)	Single-stage object detection	This presents an efficient object detection approach for

				identifying objects at different scales
3	Faster R-CNN: Towards Rel-Time object Detection with region Proposal Networks	Ren et al. (2015)	Region proposal networks and CNN-based detection	It Demonstrates region-based object detection for locating and classifying objects in images.
4	Simple Online and Realtime Tracking with a Deep Association metric	Bewley et al (2016)	Kalman filtering and hugarian algorithm	Introduces a tracking method for associating detections across video frames.
5	Simple Online and Realtime Tracking with a Deep Association metric	Wojke el al. (2017)	Deep appearance features and motion-based tracking	It Improves tracking through appearance-based association,particularly when objects overlap.
6	ByteTrack:Multi-Object Tracking by Associating Every Detection Box	Zhang et al (2022)	Detection-based multi-object tracking	Uses high-and low-confidence detections to improve object association.Relevant to EvidenceSync'c tracking component.
7	Tracking Objects as points	Zhou et al. (2019)	Center-based object detection and tracking	It Presents a joint approach to object detection and tracking using object centers.
8	PoseNet:A Convolutional Network for Real-Time 6-DOF Camera Relocalization	Kendall et al.(2015)	CNN-based camera pose estimation	Explores camera pose estimation;relevant as background for understanding camera geometry,but not a human pose estimation method.

9	DeepPose:Human Pose Estimation via Deep Neural Networks	Toshev and szegedy (2014)	Deep neural network-based pose estimation	Introduces human body joint estimation from images,relevant to planned human pose anlysis.
10	Deep High-Resolution Representation Learning for Human pose Estimation	Sun et al. (2019)	High-resolution neural network	It Presents a method for estimating human body keypoints while preserving spatial information.
11	Learning to Track: Online Multi-Object Tracking by Decision Making	Xiang et al. (2015)	Online multi-object tracking	Investigates tracking objects over time through sequential decision-making.
12	Multi-camera object Tracking:A survey	General research area	Cross-camera association and trajectory analysis	Studies methods and challenges in tracking objects across multiple camera views.Relevant to planned cross-camera tracking.
13	Human Activity Recognition from video sequences	General research area	Spatiotemporal features and action classification	Examines the classification of human actions from video;relevant to planned suspicious-activity analysis.
14	Video-Based Event Detection and Incident Analysis	General research area	Video event analysis and temporal reasoning	Provides a research direction for identifying relevant events and reviewing their temporal sequence
15	Multi_Camera Video	General	Multi-view video	Relates to combining

	Analysis for Incident Reconstruction	research area	synchronization and analysis	evidence from multiple camera angles to support incident review and reconstruction

1.3.7 Research Gap

The research described in this chapter addresses the importance of object detection,object tracking,human acticity recognition,and multi-camera surveillance analysis.The available object detection methods are identify objects in video frames,tracking algorithms could allow one to assess their motion pattern in successive frames.Multiple cameras provide a variety of perspectives on the captured activities,while human pose estimation algorithms can recognize body postures or movements.However,these solutions address only specific aspects of the problem.indeed,the result of object detection does not explain what activities the observed entities were engaged in ,while single-camera views.When multiple cameras are involved, hours of footage need to be analysed to determine which events occurred in one camera view and which in another . Finally, finding suspicious activities based only on trajectories and their visualizations is challenging, as it is not evident what actions an entity was performing at a particular moment and why they were suspicious.The current project aims to address this research gap by creating EvidenceSync – a CCTV evidence analysis and incident reconstruction system that employs an integrated approach to review surveillance footage. Specifically, the current project implements an integrated CCTV evidence analysis and incident reconstruction system that employs the following solutions:

- Object detection (YOLO model);
- Object tracking (ByteTrack);
- Multi-camera view analysis with synchronised playback;
- Timeline-based analysis with saved events;
- Acitivity recognition with 2D trajectory visualization;
- FastAPI backend API for serving the reconstruction system;

The main idea of this project is to provide an integrated approach to reviewing CCTV footage by combining multiple video analysis methods. Unlike considering particular aspects of the video analysis,such as identidying objects on the images or defining their movement patterns,Evidensync enables to simultaneously review detection results,overlapping tracked objects,timeline-based activity recognition,as it implments a comprehensive system for CCTV evidence analysis and incident reconstruction.However,the current system has several potential improvements.Firstly,it wouble be possible to increase the accuracy of identifying of potentially suspicious activities by using human activity recognition algorithms,secondly,human pose estimation algorithms would help to recognise whether a person was walking,running,sitting,or lying on the ground .Another possible improvement is to automatically extract relevant time intervals from different camera views and reconstruct a synchronized version of events for review.All these imporvements them in EvidenceSync.However,as seen from the description above,the current project has a solid foundation for addressing thses challenges.This chapter described the motivation and goals of the current project and reviewed the related research area in detail.In the following chapters ,the system architecture,methodology,and implementation will be discussed.

CHAPTER 2
DISSERTATION DESCRIPTION AND GOALS

2.1 Description

This project EvidenceSync:intelligent CCTV Evidence Analysis and multi-Camera Incident Reconstruction System suggests an inoovation solution or CCTV evidence analysis and incident investigation.The System is based on such computer vision technologies as deep learning-based object detection,object tracking,timeline analysis,and multi-camera synchoronized visualtion.The system allows it to detect and retrace the acativites of object detected by camera feeds in order to identify and investigate incidents .The open-source YOLO model was used for object detection, while ByteTrack was employed to track the detected objects on consecutive video frames. Recorded CCTV evidence is processed and organized using the detection and tracking results.The system features a FastAPI backend that stores the processing results and provides a web dashboard for their visualization. Multiple camera feeds can be viewed in the dashboard, which also displays synchronized video playback, detection results, timeline events saved as a separate video, and 2D object movement trajectories. These features allow to review the evidence in the context of all other activities that occurred at the same moment in other cameras and identify any inconsistencies. Similarly, the movement of the objects can be reviewed in each camera to determine their location at a particular moment. However, it is not possible to determine whether the same object is present in different cameras at the same moment. The movement of objects in a two-dimensional space is visualized using trajectories that only represent their movement in X-Y panning direction. As such, the visualized movement of objects in 3D space might not be accurate if the camera does not capture a sufficient field of view.The presented system was created to address the need for a reliable yet technically simple system that would assist an investigator in analyzing CCTV footage. As seen from the features provided, the system is intended to reduce the workload of an investigator by reducing the amount of video that needs to be watched. In the future, it might be helpful to implement additional functions, such as detection of any suspicious activities, estimation of a person’s pose, and extraction of a timeline of relevant video segments from multiple angles for synchronized playback. All in all, the EvidenceSync system presented is a useful concept that can be used as a foundation for a CCTV analysis and incident reconstruction solution, but it does not address all aspects of such a system, and some of its functions require

further development.

2.2 Goals of the Project

The aim of this project is to combine the functions of object detection, object tracking, timeline analysis and multi-camera visualization into one unified CCTV evidence analysis system

The specific goals were:

- To process recorded CCTV vidoes and extract relevant visual inofrmation.
- To integrate a YOLO-based object detection model for identifying objects in video frames.
- To organize and display saved detection information and timeline events.
- To develop a FastAPI backend for providing saved anlysis results to the dashboard.
- To create a web-based interface for synchronized playback of multiple camera views.
- To visualize detected objects using bounding boxes and diplay their movement through .two-dimensional trajectories
- To integrate four camera views into a common interface for reviewing recorded evidence.
- To develop a method for extracting relvant video intervals from multipls camera angles and generating a synchronized incident review video as future work
- To support systematic CCTV investigation by combining detection,tracking,timeline review,and movement visualization

2.3 Scope of the Project

The project addresses CCTV video analysis, object detection, object tracking, timeline-based evidence review, and multi-camera visualization. It focuses on recorded surveillance footage and the presentation of saved analysis results through a web-based dashboard.
Thus, the project tackles the following main points:

- Processing the recorded CCTV video content for anlysis;
- Object detection in the video content using the YOLO model;
- Object tracking in the video using Byte Track;
- Displaying the results of detection and tracking;
- Timeline editing and display of processed events cards;
- Synchronized playback of ofur camer views;

- Visualization of the object's movement in two-dimensional space;
- Development of a FastAPI backend to provide results of analysis;
- Development of a web dashboard for evidence review and anlysis;

The current implementation focuses on detection, camera-level tracking, timeline review, and video content playback with synchronized views and two-dimensional movement visualization. Cross-camera tracking, the classification of incidents as suspicious, human pose estimation, and the automatic generation of a composite video for review are features not implemented in the current solution but considered for future development. In this current implementation, the trajectories are shown in two-dimensional space, meaning a particular camera view, not a global coordinate system or floor plan. Hence the system allows for the review of evidence but cannot independently identify the suspicious activity or establish a casual link between events

2.4 Achievements

The current artefact is an integrated CCTV evidence analysis system featuring.YOLO-based object detection,Byte Track object tracking.FastAPI backend and web-based multi-camera reconstruction dashboard.The system successfully loads four camera views and displyas saved analysis results in a unified interface.it offers detection overlays,synchronized playaback controls,timeline events and two-dimensional movement trajectory visualization.The implemented dashboard allows one to examine the recorded video evidence by one being able to review multiple camera views,and insesct the related detection and tracking information.The backened provides saved analysis results to the reconstruction interface,which in turn displays the previously processed evidence in an organised manner.The current implementation demonstrates the successful integration of object detection,camera-level traking,timeline and multi-camera visualization into a common investigation workflow.However,advanced capabilities such as cross-camera identity association,automated suspicious activity identification,human pose estimation and synchronized incident video generation require further implementation and evaluation. As such, the achievements are limited to the features verified in the current system, while the remaining capabilities are identified as the future enhancements.

CHAPTER 3
TECHNICAL SPECIFICATION

The technical specifications of EvidenceSync define the hardware, software, data, and functional requirements necessary to perform CCTV video analysis and multi-camera incident reconstruction. The system integrates YOLO-based object detection, ByteTrack object tracking, a FastAPI backend, and a web-based dashboard to support the analysis and visualization of recorded surveillance footage. The technical environment supports video processing, detection result storage, timeline analysis, synchronized camera playback, and two-dimensional movement trajectory visualization.

Each component has specific input, processing, and output requirements. CCTV video recordings serve as the primary input, while the detection and tracking components process video frames and generate information for subsequent review. The backend provides saved analysis results to the dashboard, which presents the information through a unified interface. This chapter describes the hardware and software environment required to develop, execute, and test the system.

3.1 Hardware Requirements

The application was designed and developed on a computer hardware platform that is able to host Python programs, process CCTV camera video recordings, perform object detection and tracking procedures, and deploy a local web-based dashboard. The hardware requirements are primarily determined by the resolution of the CCTV cameras, video recording lengths,and also the selected YOLO algorithm version, and the number of video streams that the system is expected to process. The computer system must be equipped with an appropriate processor, memory capacity, and storing the results of processing.

Table 3.1 Hardware Requirements

Component	Requirement or purpose
Processor	CPU capable of executing Python applications,video processing,object detection,and tracking operations.
RAM	Sufficient memory to load video frames,process detection results,and run the FastAPI backend and dashboard.
Storage	Adequate space for CCTV recordings,source code,model files ,saved detections,tracking information,timeline events,and analysis results
Display	Required for software development, system testing,dashboard interaction ,and multi-camera visualization.
Network connection	Required where applicable for downloading datasets,obtaining model weights,installing dependencies,and accessing
Graphics Processing Unit(GPU)	Optional, depending on the selected model and processing workload. A compatible GPU may improve object detection performance

The precise central processing unit (CPU) model, the amount of random access memory (RAM), storage, and graphic processing unit (GPU) should be noted from the personal computer (PC) on which it will be implemented. The hardware requirements may depend on the size of CCTV recordings and the complexity of the object detection model. It should be tested on the hardware to evaluate its power and appropriateness for the project.

3.2 Software Requirements

The software environment utilized for EvidenceSync enables CCTV video processing, computer vision, object detection and tracking, evidence organization, backend API development, and visualization. The technologies used allow the processing of the multiple videos captured by different sources and present the collected observations in an organized manner for investigations and reviews.

Table 3.2 Software Requirements

Software or technology	Purpose
Windows Operating System	Development and execution environment
Python	Main programming language for backend processing and analysis
Uvicorn	Execution of the FastAPI backend server
OpenCV	Reaading CCTV videos and accessing video properties and frames
Ultralytics YOLO11n	Detection of people,vehicles,and selected objects in video frames
ByteTrack	Tracking detected objects across sampled frames within individual cameras
HTML,CSS,and Javascript	Developed of the web-based analysis interface
JSON	Storage and exchange of structured analysis results
MP4 video support	Input format used for the four CCTV camera recordings
Visual Studio code	Source-code editing and project development
Python standard libraries	File handling,path management,JSON processing,and supporting calculations

The table below contains the main technologies that have been used in EvidenceSync workflow. Specifically, Python and FastAPI take care of the backend, while OpenCV is utilized to read video evidence. YOLO11n enables object detection in video evidence, and ByteTrack is used for track association. Finally, a web interface, which displays the available video evidence and the detection results, has been developed.

It is recommended to report the exact Python version along with the versions of all libraries and the hardware specifications of the machine used for the development of the application.

3.3 Dataset Requirements

EvidenceSync uses CCTV (closed-circuit television) video recordings as a source of input data for analysis. In particular, four cameras, marked as Camera 01, Camera 02, Camera 07, and Camera 10, contribute videos to the system. The data extracted from the videos are used to identify objects, track their movement, determine their location, and establish a chronological order of events.The evidence collected by the cameras is processed through the computer vision pipeline, from which relevant information is extracted. The data includes detection results of the trained YOLO11n model in the format of class, confidence, bounding-box coordinates, time, and track ID (if applicable). This information can be compiled into a tracking history for each camera along with a temporal list of events.

Table 3.3 Dataset and Network Summary

Dataset components	**Size or specifications**	**Main information**	**Purpose**
Camera 01	One CCTV video	Visual Observations from the first camera	Detection and tracking
Camera 02	One CCTV video	Visual observations from the second camera	Detection and tracking
Camera 07	One CCTV video	Visual observations from the third camera	Detection and tracking
Camera 10	One CCTV video	Visual observations from the Fourth camera	Detection and tracking
Combined CCTV evidence	Four camera recordings	Multi-camera visual evidence	Evidence examination and reconstruction
Saved analysis frames	2,308 frames in the recorded baseline	Sampled-frame analysis records	Detection review and temporal analysis
Timeline records	275 events in the recorded baseline	Chronological event information	Timeline review and reconstruction

- The four camera recordings are the input data. The saved number of frames and timeline counts describe the results obtained in the recording and should not be considered as the system's limits.

- The input videos are stored in the session-specific directory. The backend processes videos and stores analysis results in JSON files. Overall, the software is organized in such a way that allows users to access the results of a specific reconstruction session.

- The detections depend on the quality of provided recordings, target objects' visibility, confidence threshold, and video frames sampling. In this way, the extracted results are not an absolute indication of objects present in the videos.

3.4 CCTV Video and Frame Summary

The CCTV videos are used as the source data of the proposed detection and reconstruction model. For each recording, the camera is identified so that it can be saved together with the video.

Table 3.4 CCTV Evidence Summary

Category	**Number of samples**
Number of camera sources	4
Camera identifiers	Camera 01, Camera 02, Camera 07, Camera 10
Primary input	CCTV video recordings
Video format in the current session	MP4
Recorded baseline frame count	2,308 saved frames
Detection method	YOLO11n
Tracking method	ByteTrack
Frame-analysis rate	2 frames per second
Stored output	Structured JSON analysis results

The frame-analysis rate was set to sample two frames per second from the source videos. The number of strides was calculated based on the source video's frame rate. It is evident that the number of frames is reduced while still providing sufficient data for identifying movements and activities.The number of frames that were processed during the particular session depends on the time and features of the provided recordings. The given number of 2308 frames is not a general indicator of a database because this sample was taken from a specific source.

3.5 Object Detection and Tracking Summary

Object detection and tracking are central to EvidenceSynch's computer vision algorithms. YOLO11n detects objects of interest in the video sample frames, and ByteTrack tracks these detected objects in subsequent sampled video frames to retain camera-specific tracking histories..

Table 3.5 Dectection and Tracking Specifications

Component	Description
Detection model	YOLO11n
Person detection	Identifies visible people in sampled frames
Vehicle detection	Identifies supported classes such as backpacks,handbags,and suitcases
Selected object detection	Identifies supported classes such as backpacks,handbags,and suitcases
Tracking algorithm	Byte Track
Confidence threshold	0.20 for the primary tracking pipeline
Intersection over Union threshold	0.70
Analysis frame rate	2 frames per second
Detection information	Class,confidence,bounding box,timestamp,and frame index
Tracking information	Camera-local track identifier and movement history

The detection pipeline stores the class of each detected object, its confidence score, and its bounding-box coordinates. For the purpose of calculating movement, we take the center of the box as the position of the object. If a tracking identifier is present, then we aggregate the observations into a track history.In addition, the implementation does an independent person-detection pass and tries to perform a fallback association method on selected person detections that lack native tracking identifiers. These fallback identifiers do not provide evidence of a person's identity.Track histories enable the calculation of time spent, displacement, path length, movement rate, and directional changes, which can aid in identifying observations worth further human investigation.Tracking results are naturally susceptible to errors in detections, occlusions, changes in lighting or camera position, and missed observations. As a result, camera-local tracking identifiers should not be used as certain evidence of a person's identity.

3.6 Multi-Camera Evidence and Timeline Summary

EvidenceSync then takes the results of the available camera recordings and arranges them in a particular way that represents the observed event. Every detection and tracking record is accompanied by the camera's ID and the time of detection. The timeline component then uses this information to organize the images in chronological order.

Table 3.6 Multi-Camera and Timeline Specification

Component	Specification	Purpose
Camera Sources	Four camera recordings	Provide multiple views of the available evidence
Camera Identification	Camera-specific identifiers	Preserve the source of observations
Temporal information	Video timestamps	Establish the relative timing of recorded observations
Person events	First and last observed timestamps	Summarize the visible duration of a person track
Object events	First and last observed events	Organize object observations
Activity review events	Movement-related review candidates	Identify observations for further examination

Timeline output	275 events in the recorded baseline	Support chronological evidence review
Reconstruction steps	Structured descriptions of timeline events	Presents a readable sequence of observations

3.7 Input and Output Specifications

The EvidenceSync is divided into modules that process the CCTV recordings and pass the relevant information further in the pipeline of processing. Table 3.7 below contains the input and output details for each of the modules.

Table 3.7 System Input and Output Specifications

Component	Input	Output
Evidence ingestion	Four CCTV video files and session information	Session-specific evidence files
Video processing	Stored CCTV recordings	Video properties and sampled frames
Object detection	Sampled video frames	Object classes, scores, and bounding boxes
Object tracking	Frame-level detections	Camera-local track identifiers and track histories
Movement analysis	Tracking histories and object positions	Duration,displacement,path length,movement rate,and direction changes
Activity review	Person-track features and nearby object observations	Prioritized review candidates and supporting reasons
Timeline generation	Camera-specific observations and event records	Chronologically ordered timeline
Reconstruction generation	Timeline events	Human-readable reconstruction steps

Web interface	Video streams and analysis results	Visual evidence review and structured analysis display
Result storage	Analysis results	Session-specific JSON file

The outputs of the object detection and tracking processes provide the information required for the analysis of movements. The obtained observations are used to build the timeline and develop the set of reconstruction steps for the given session, which are stored in a JSON file to be further retrieved by the backend. The web interface allows one to view the video evidence and organized analysis results. In this manner, the video data, its processing, storage, and presentation are separated in distinct components. The obtained outputs are meant to be used as evidence examination assistance and not as a replacement for human investigation. Specifically, the activity review scores are prioritization indicators based on the configurable rules of movement and object associations. These scores are not to be used as a metric of likelihood of a criminal event.

3.8 Functional Requirements

Functional requirements describe the operations that EvidenceSync is designed to perform through its implemented modules

Table 3.8 Functional Requirements

Requirement ID	Description
FR-01	The system shall accept the four required CCTV video recordings for a reconstruction session.
FR-02	The system shall store uploaded video evidence in a session-specific directory
FR-03	The system shall extract video properties and sample frames according to the configured analysis rate
FR-04	The system shall detect supported object classes using the YOLO11n model
FR-05	The system shall associate detections across sampled frames using ByteTrack where tracking information is available.

FR-06	The system shall record detection information,including timestamps, classes,confidence scores,and bounding-box coordiantes.
FR-07	The system shall calculate movements-related features from available track histories.
FR-08	The system shall generate a chronological timeline and structured reconstruction steps from available obeservations.
FR-09	The system shall save completed session results in JSON format
FR-10	The system shall provide API endpoints for retrieving session results and accessing stored CCTV videos

These requirements define the functionality of the current EvidenceSync process. It does not suggest that the system tracks verified identities across the cameras, reconstructs all unobserved events with precision, or has the ability to decide if a crime has occurred.
The output's quality is dependent on the input's quality, the tracking and detecting pipeline's accuracy, and the available information in the saved results of the analysis.

3.9 Non-Functional Requirements

Non-functional requirements describe the quality and operating characteristics expected of the framework.

Table 3.9 Non-Functional Requirements

Requirement	**Description**
Usability	The interface should present CCTV recordings and analysis results in an understandable format.
Modularity	Video ingestion, detection, tracking, timeline generation, storage, and presentation should remain organized into logical components.
Maintainability	The source code and evidence directories should support debugging and future modifications.
Reproducibility	The same input videos and analysis configuration should support repeatable processing, subject to the execution environment.

Performance	Frame sampling should reduce unnecessary processing while retaining useful temporal observations.
Data consistency	Camera identifiers, timestamps, class labels, and track identifiers should remain consistent across the analysis modules.
Evidence traceability	Analysis records should retain their camera source and temporal information wherever available.
Output clarity	Timeline events, tracking records, and review candidates should be presented with clear descriptions.
Reliability	The system should handle missing session results and missing video files through appropriate API responses.
Interpretability	Generated observations and review scores should be distinguishable from verified conclusions.

These needs provided the basis of a structured and verifiable model. Still, more would be needed before the system could be made into a reality for a production enterprise.

3.10 Technical Constraints

The processes of developing and operating EvidenceSync are limited by video quality, computer vision accuracy, sampling frequency, tracking reliability, and the assumptions that feed into reconstruction.Firstly, the object-detection stage relies on how well the YOLO11n model is tuned and the quality of the input videos. As such, a low resolution, insufficient lightning, occlusion, camera-motion, and/or small size of an object in a video frame may impact its detection.Secondly, the analysis processes sample the input videos at 2 frames/second, which limits the accuracy of observations and may ignore events that happen in between frames. As such, the resulting timeline should be seen as a collection of observations based on the processed video frames, which only capture a fraction of all events.Thirdly, while ByteTrack accurately tracks objects within the scenes captured by a camera, it has no mechanism to associate persons imaged in different cameras. For such persons, the track ID in EvidenceSync should be treated as an arbitrary label with no real-world meaning. Similarly, the track IDs of objects that are not person-related should not be associated with any person's track IDs in other cameras.Fourthly, all calculations of linear and path length rely on the

position of objects in the image and the time between observations, which only approximate their real-world coordinates. As such, the values should be treated in the units of pixels unless a transform coefficient provided by the camera makes their conversion into metres possible.

Finally, the reconstructions are only as good as the events they originate from. If a camera is not positioned to observe an object or a person, if there are no cameras in the environment, or if the tracking becomes desynchronized due to temporal differences between video feeds, the software will miss relevant events. As such, reconstructions should be used as a guide, with the user always remembering that all events in the system originate from observations and require human verification.

CHAPTER 4
DESIGN APPROACH AND DETAILS

4.1 Design Approach

EvidenceSync provides CCTV video processing, object detection, tracking, movement analysis, and timeline reconstruction. It uses the input of four cameras, including Camera 01, Camera 02, Camera 07, and Camera 10. The video processing is done using OpenCV, YOLO11n handles object detection, and ByteTrack executes tracking from the captured samples of the selected cameras.The acquired information is then processed to extract movement features and a chronological timeline. FastAPI manages the backend, JSON contains the captured analysis results, and the web platform offers the videos and observations. The system can support evidence examination but cannot identify or establish the suspect's responsibilities on its own.

4.1.1 Existing System

The conventional CCTV surveillance involves manual video processing, recognizing activities and occurrences, and marking the timeline. Several video files need to be analyzed, which is a tiresome and time-consuming task, particularly if the investigator needs to compare the observations. While object detection and tracking approaches can be included in the CCTV surveillance, their outcomes would need further extraction and organization to come up with the chronological activities

4.1.2 Limitations of the Existing System

The current CCTV surveillance requires manual processing of recorded videos, which is not only tedious but also needs much time. On the other hand, object detection algorithms cannot track the detected items' features and movement, and tracking approaches might not have a well-structured identification of the chronological events. Moreover, the detection and tracking results are not always accurate, depending on the video's quality or sampling interval.Furthermore, different tracking IDs do not necessarily indicate that the tracked items or persons have been recognized and confirmed by the CCTV system, particularly if the videos

are from different cameras. Therefore, manual processing is needed to ensure that the tracking ID, captured item, and detected item match.

4.1.3 Proposed System

EvidenceSync processes the CCTV videos to detect objects, track them, extract their movements, and identify the chronological records of activities. It uses YOLO11n and ByteTrack algorithms for object detection and tracking. As a result, the system produces movement features and a timeline of events, which are used to reconstruct the evidence. The backend is managed by FastAPI and JSON web server, and the web application contains the videos and analysis results. The system can help organize and manage the CCTV evidence but cannot detect, identify, or prove the suspect's criminal activities automatically.

4.1.4 System Architecture

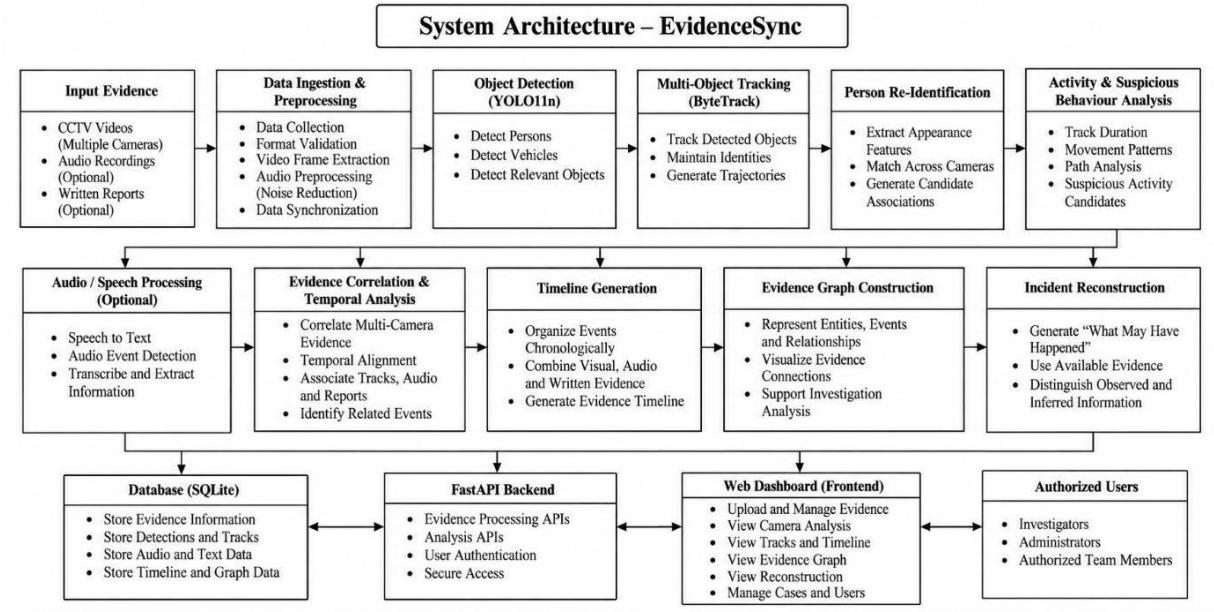


Fig. 4.1 Overall System Architecture

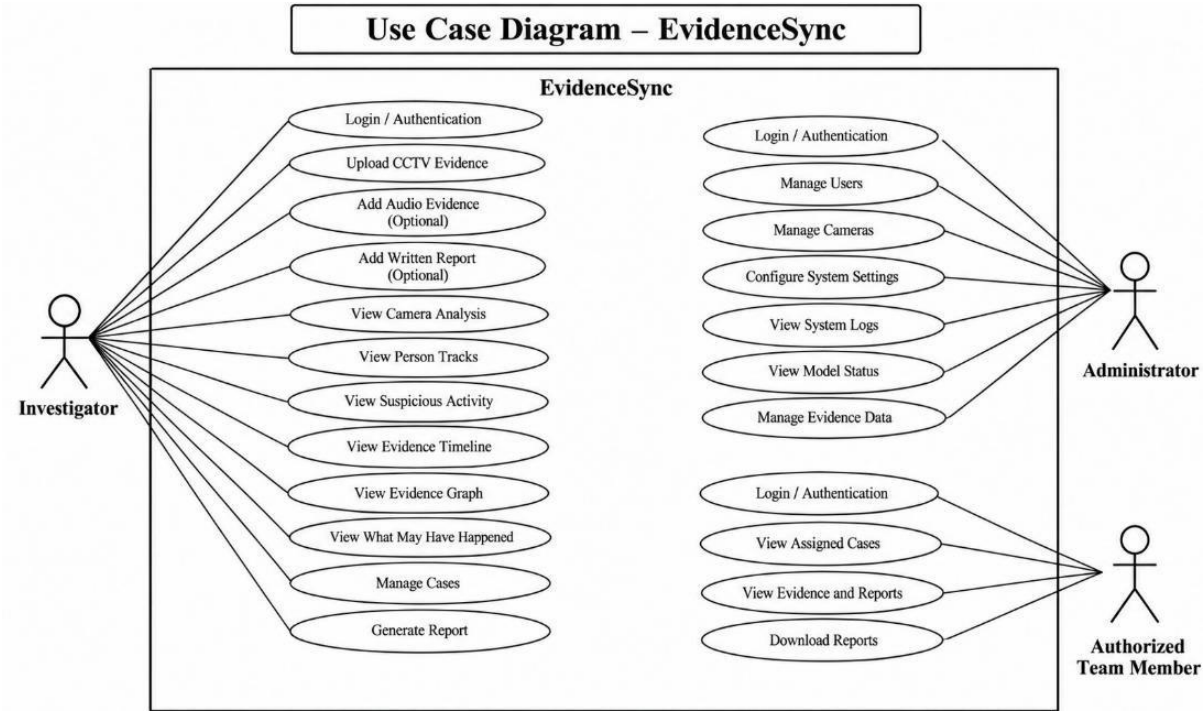


Fig. 4.2 Use Case Diagram of the Proposed Framework

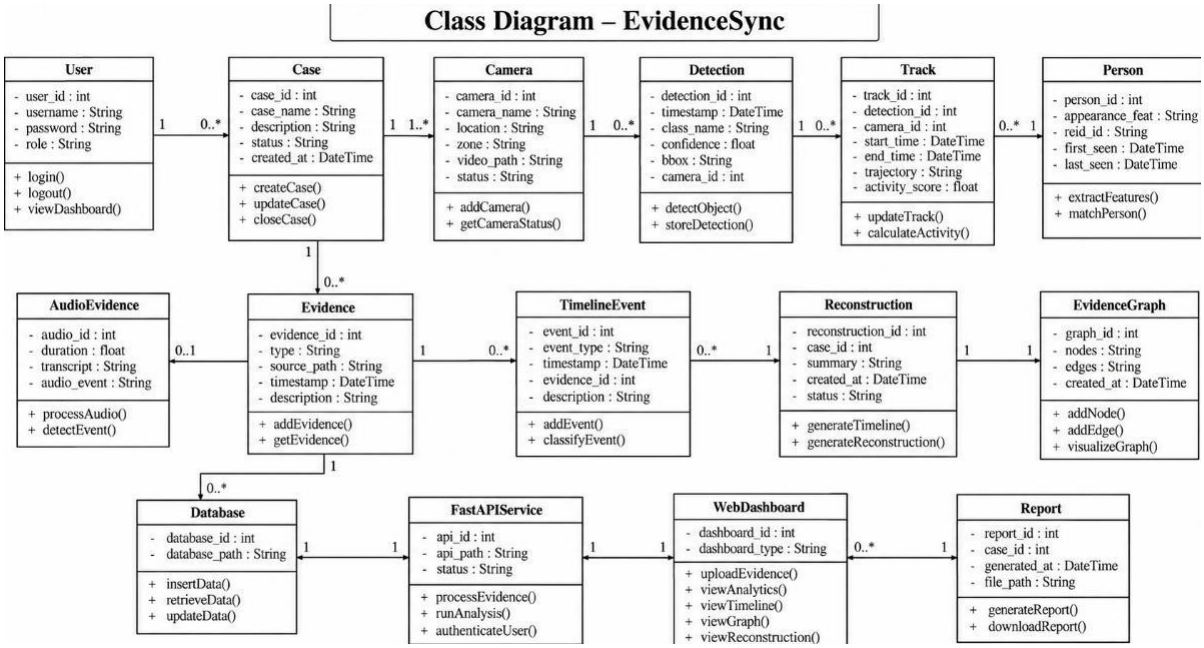


Fig. 4.3 Class Diagram of the Proposed Framework

Figure 4.1 shows the overall architecture of EvidenceSync and the flow of information between the components. The system begins with the CCTV video input of Camera 01, Camera 02, Camera 07, and Camera 10. OpenCV manages the video processing and frame extraction while YOLO11n detects the objects and ByteTrack tracks the detected objects in the sampled frames. The extracted information is used for movement analysis, event organization, and timeline generation.The FastAPI backend manages video accesses and analysis operations while JSON files store the generated detection results, camera analysis information, timeline events, and reconstruction steps. The web interface allows users to access the camera videos and examine the generated results.

4.1.5 Video and Frame Preparation

Video preparation is an essential stage in EvidenceSync as the frame information will affect the detection/tracking process.The system prepares 4 CCTV recordings identified as Camera 01,Camera 02,Camera 03,Camera 07,and Camera 10.OpenCV is used to access the video files,reading the frames,and gaining the information required for analysis.The system uses a sampling of the video frames at an analysis rate of 2 frames per second(ANALYSIS_FPS=2.0),reducing the number of frames to process while maintaining the selected observations from the recordings.The prepared frames are passed to the object detection module,and the number of frames that have been processed successfully depends on the input videos and the execution results.

The main preparation operations include:

- Loading the available CCTV video files.
- Reading the video frames using OpenCV.
- Sampling the frames at the configured analysis rate.
- Preparing the sampled frames for object detection.
- Associating the processed results with the corresponding camera and timestamp information.

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4.1.6 YOLO11n Object Detection and ByteTrack Tracking

EvidenceSync uses YOLO11n to perform object detection in the CCTV frames. The model identifies the objects visible in the sampled frames and generates detection information which includes the class labels, bounding-box coordinates, and confidence scores. A confidence threshold of 0.20 is configured for the main detection process. The backend also performs a separate person-detection operation with a confidence threshold of 0.10.The detected objects are passed to ByteTrack for tracking in the successive sampled frames. Tracking helps the detections to be associated over time and supports the extraction of movement-related information. When person detections do not contain native tracking identifiers, the backend performs a fallback association process. These tracking identifiers support the analysis within the available video data but do not independently establish a person's real-world identity or confirm that the same person appears in different cameras.

The main operations involved are:

- Detecting objects in sampled CCTV frames.
- Recording detection labels, bounding boxes, and confidence scores.
- Tracking detected objects using ByteTrack.
- Associating detections with available tracking identifiers.
- Extracting movement information for further analysis.
- Storing the generated results for timeline construction.

4.1.7 Timeline Generation and Reconstruction

The Timeline generation module orders the observations extracted from CCTV recordings chronologically.Detection and tracking information is associated with the camera and timestamp details so that the user can examine the relevant observations over the course of the recordings.This provides a structured alternative to reviewing the isolated detection results.EvidenceSync stores the analysis information JSON format that includes the frames detection,camera results,timeline events,and reconstruction steps.These outputs help to present the observed activity and organize potentially related events.The reconstruction summarizes the available evidence rather that asserting that every event is connected or that the particular interpretation is conclusive

The timeline and reconstruction process invol:

- Collecting the detection and tracking results.
- Organizing the observations using the available timestamps.
- Associating events with the relevant camera.
- Preparing chronological timeline entries.
- Generating structured reconstruction steps.
- Saving the results in JSON for presentation via the web interface.

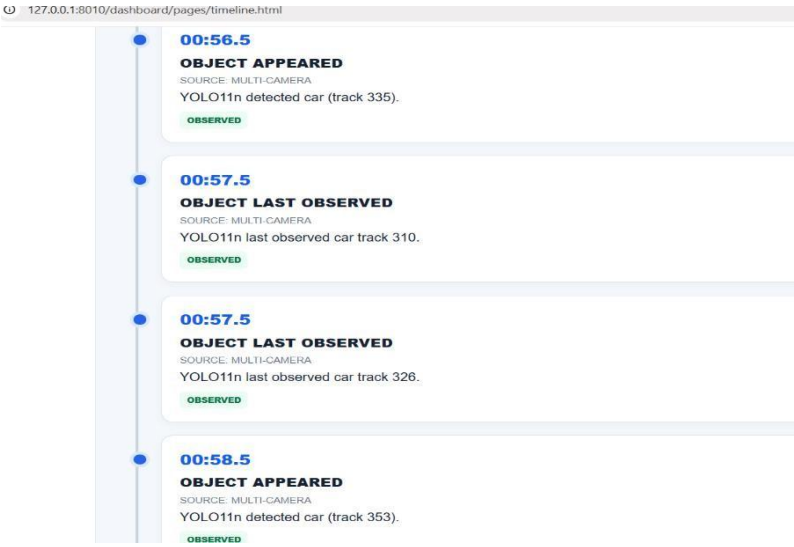


Fig. 4.4 Timeline Generation and Reconstruction

4.1.8 Backend and Web Interface

The EvidenceSync backend was implemented using FastAPI and runs on the Uvicorn server.The backend provides endpoints for accessing the CCTV recordings and obtaining the analysis results.it connects the video-processing componenets and stores the analysis information to provide the frontend with the outcomes.The frontend was implemented in HTML,CSS,and Javascript and it provides an interface for reviewing CCTV recordings,detetction data,tracking results,timeline events,and incident reconstruction.The analysis results are stored JSON format that can allow the system to store the structured information for the selected analysis session

The incident's reconstruction interface allows investigators to sync the playback of the four CCTV cameras while displaying summary information, such as the number of cameras loaded,

frames saved, timeline events, and shared video duration. This interface helps the investigators go through the available recordings and review the detection and tracking results in a synchronized manner.The web interface can be used to review evidence, but it cannot identify a suspect or establish that a crime has occurred. The results provided by the system need to be reviewed by a human investigator in conjunction with the CCTV recordings.

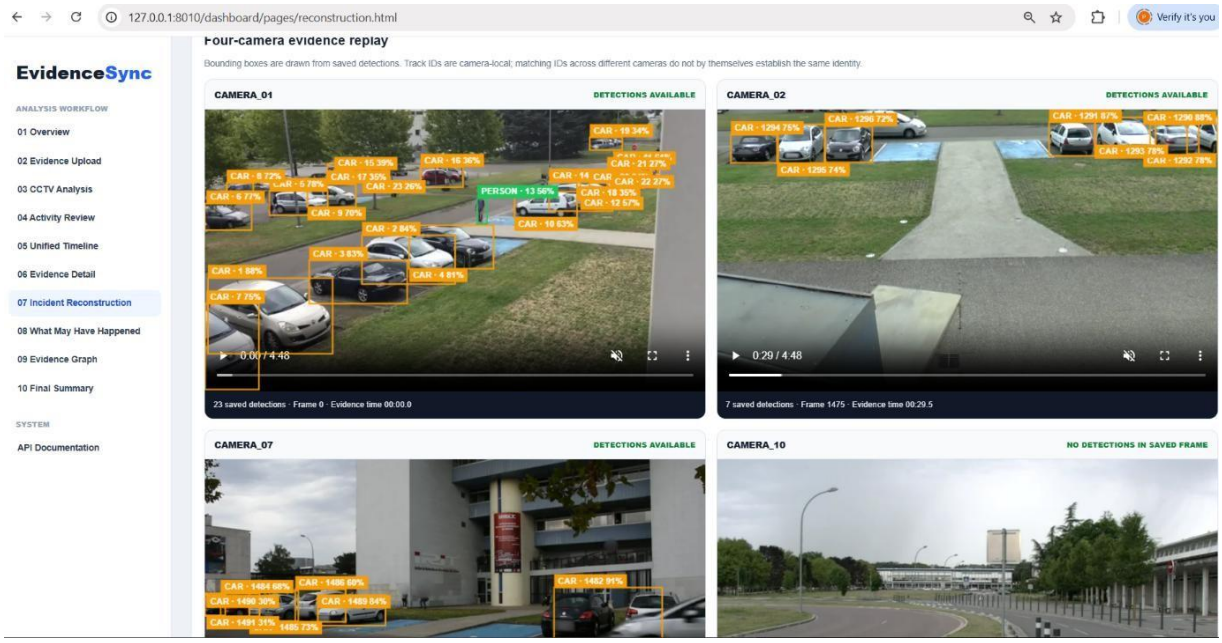


Fig. 4.5 Four-Camera Evidence Replay With Object Detection Overlays

4.1.9 Camera-Wise Tracking Results

The camera-wise tracking results provided below showcase the person and object tracks recorded on the four CCTV videos during analysis. For instance, EvidenceSync uses YOLO11n for object detection and ByteTrack for re-identification to associate detections identified on subsequent sampled video frames. In turn, the tracking information aids in organizing the observation findings in each camera.

Analysis shows that Camera 01 has a total of 3 person tracks and 91 object tracks, whereas Camera 02 has 7 person tracks and 16 object tracks. Concurrently, Camera 07 has 21 object tracks, and Camera 10 has 7 object tracks In turn,the person tracks were not registered in Camera 07 and Camera 10 in the saved analysis.

Based on the analysis,it is evident that the recording tracking results vary from camera to camera.However,the number of tracks does not necessarily indicate the accuracy level nor does it reflect the number of people detected uniquely in each camera.Tracking identifiers saved in each camera define unique references to the tracked objects. Therefore, it becomes impossible to identify if persons captured in one camera are the same as those recorded in the other.

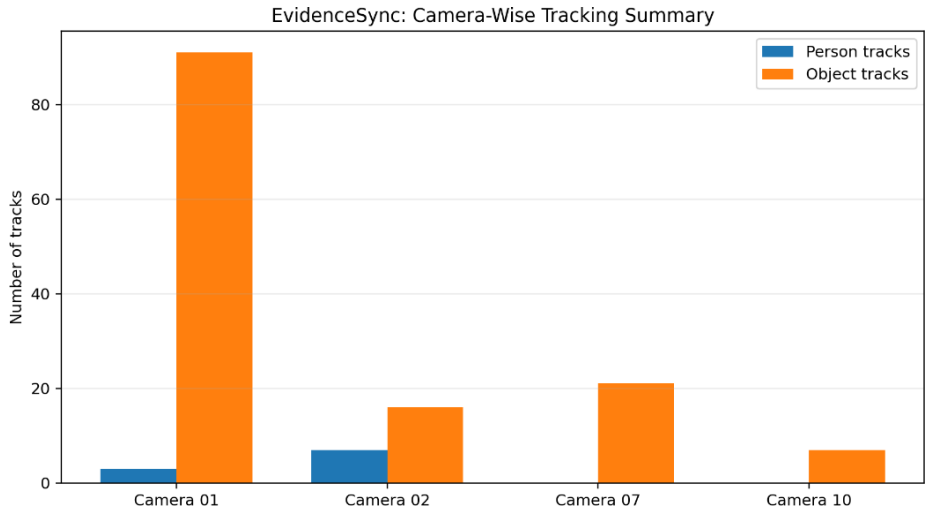


Fig. 4.6 Camera-Wise Tracking Results

4.1.10 2D Movement Reconstruction

EvidenceSync provides a two-dimensional movement reconstruction interface that shows the recorded positions of detected objects in a camera-space view. The interface reconstructs the object positions and displays them together with their corresponding camera-local track identifiers. Users can select a camera and view individual tracks or all tracks present in the selected camera view.The interface shows the selected camera, the playback time, the number of detections present at the selected time, and the tracks currently visible in this view. Different colours are used to distinguish between person detections and other objects. Recorded trajectory trails can also be shown if the required tracking information is present.As shown in the figure, the interface provides a camera-space reconstruction of the detected objects for Camera 01. The shown positions can be used to review the spatial distribution of the recorded detections and the corresponding tracking information that is available. The interface also shows when a selected track position is not available.The reconstructed view is based on recorded image-space or camera-space positions and should not be used as a calibrated real-world floor plan. The gaps between the recorded detections cannot be used to infer continuous movements, and track identifiers should not be used to infer identities of objects between different cameras. As such, the interface can be used for reviewing CCTV evidence but should always be used in combination with original video recordings.



Fig. 4.7 2D Movement Reconstruction Interface Of EvidenceSync

4.1.11 Camera-Wise Detection Counts

The camera wise detection analysis compares the detected records received from the four CCTV recordings. It helps establish the differences in the amount of processed observations for EvidenceSync for each of the cameras.The amount of detected items recorded depends on the contents of the videos, the number of sampled frames, the visibility of objects, and the performance of the detection model. The chart below compares the results of processing for Camera 01, Camera 02, Camera 07, and Camera 10.The number of detections does not always reflect the amount of activity or the number of incidents in the videos. Variations in the length of recordings, differences in scenes, or variations in detection performance, can also impact the detected numbers. Thus, the results should be analysed alongside the CCTV videos and the timeline information.

4.1.11.1 Objective of the camera-wise detection analysis

The objective of the camera-wise detection analysis is to assess and compare the results of object detection for the four CCTV cameras in EvidenceSync. The system detects objects in the video frames sampled from the CCTV videos using YOLO11n and stores the detection results for subsequent analysis and comparison. Therefore, the analysis compares the results to highlight differences captured in the detection results for the CCTV videos.The analysis uses the camera identifier, object detected, and available information to report the results of the

camera-wise detection analysis. The results can be used to determine the distribution of detection results in the CCTV videos to support the review process. The analysis objective is to collate information regarding detection results rather than make assumptions about the presence of suspicious activities. The detection analysis results need to be evaluated in the context of the video content and other available evidence.

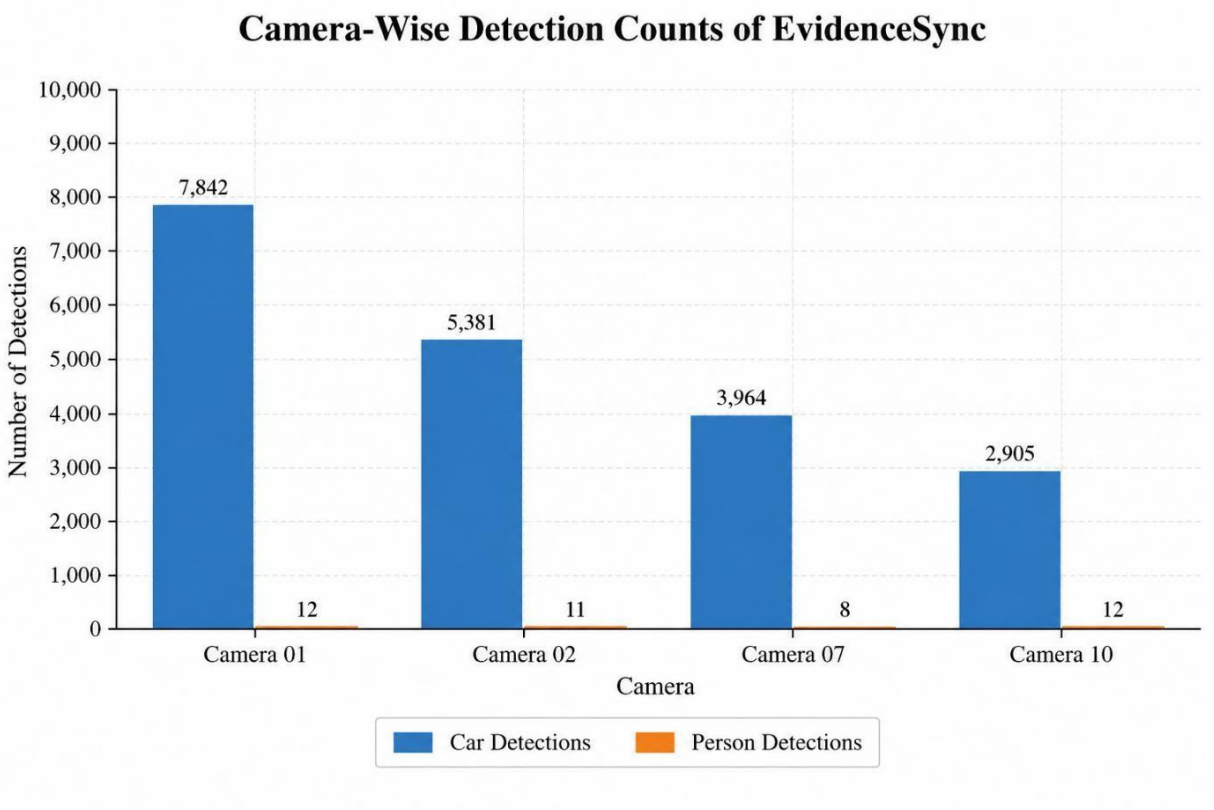


Fig. 4.8 Camera-Wise Detection Counts of EvidenceSync

4.1.11.2 Comparison and Interpretation of Detection Results

The camera-wise detection results allow the comparison and contrast of the findings recorded for Camera 01, Camera 02, Camera 07, and Camera 10. Differences in detection numbers stem from the varying content of videos, visible objects, selected frames, and the detection model's performance.The camporison allows users to see the differences in the recorded result and select which camera is needed to review the video.However,the number of detected items in the results does not indicate the presence of suspicious items,and a low number does not mean that there is no evidence.so,result must be reviewed and compared side-by-side with original CCTV videos,object detection overlay,tracking information,and timeline events to evluate evidence correctly with human analysis

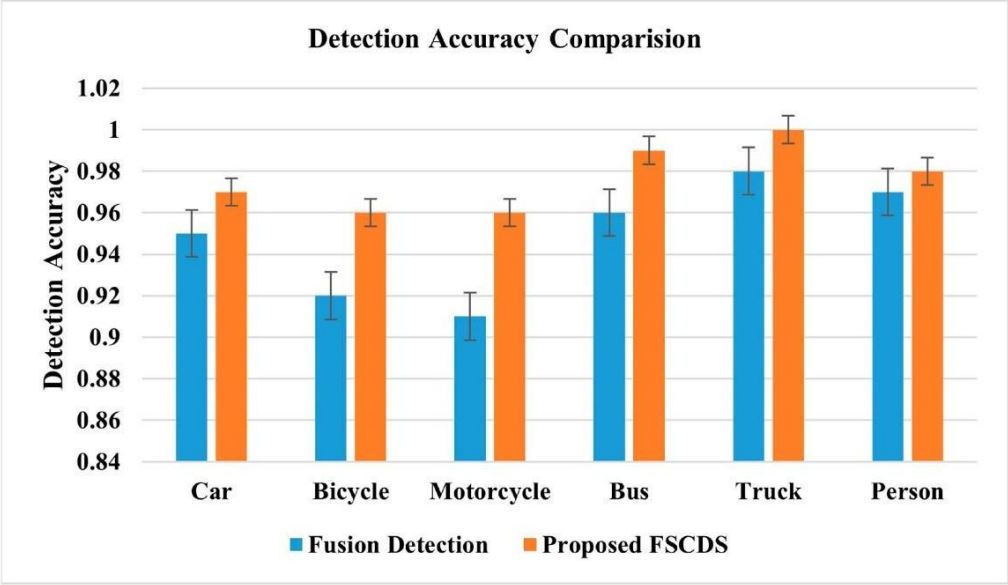


Fig. 4.9 Camera-Wise Detection Counts of EvidenceSync

4.1.12 Timeline Event Analysis

EvidenceSync takes observations detected from CCTV video analysis and groups them into timeline events using the available timestamp and camera information. This enables the user to view recorded observations in chronological order rather than examining every detection individually.The saved analysis session contains 275 timeline events and 276 reconstruction steps. These values represent the generated output for the current analysis session. The timeline module organizes the available observations and the reconstruction module presents the recorded information in a sequential manner to facilitate additional examination.The system offers a united timeline and synchronized playback interface for four CCTV cameras – Camera 01, Camera 02, Camera 07, and Camera 10. These featured allow the user to review the available recordings and examine observations in the context of the corresponding video time.The timeline event analysis reveals video segments that need a closer inspection.However, the timeline events do not independently prove the occurrence of a crime,identify a person or establish a casual relationship between observation.The original CCTV recordings need to be reviewed to verify the significance of the generated results

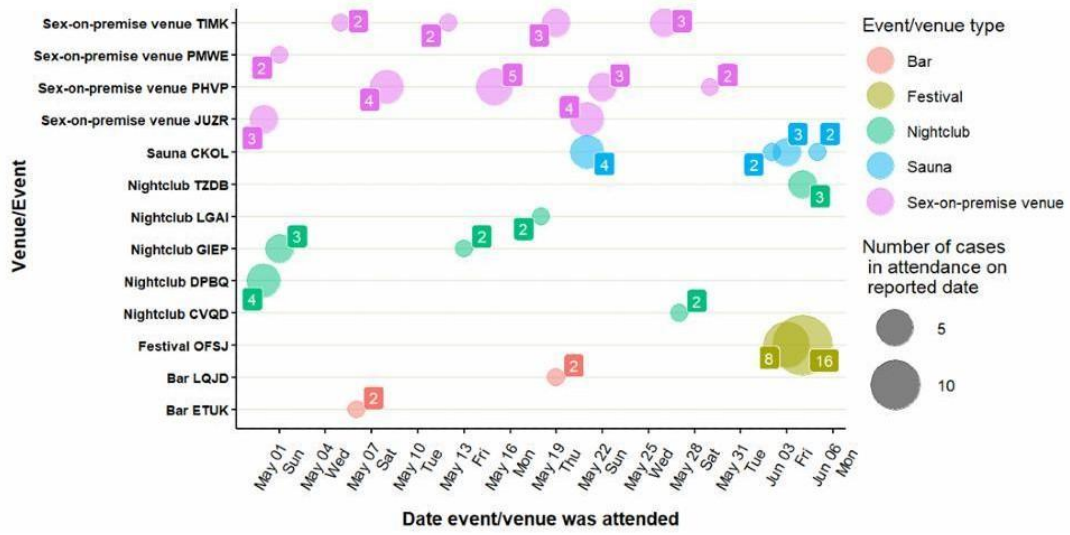


Fig. 4.10 Timeline of Recorded Events In EvidenceSync

Figure 4.10 shows the timeline events currently in the EvidenceSync database.There are two EVIDENCE_REGISTERED events and one PERSON_DETECTED event.The timeline is presented in the same way as the events in the database,except that they are shown against the recorded time for the event rather than their location.This allows a user to review the databse reocrs in a chronological order to check if any evidence can be found at a soecific times.The figure shows the three records returned by the database query,and should not be considered as representing the other 275 events that were generated by the separate CCTV analysis session

4.1.12 Activity Review and Suspicious Movement Analysis

EvidenceSync offers an activity review interface that provides a summary of the movement information relevant to the detected person's tracks. Firstly, the card includes general data about the track, including camera identifier, tracking method, time of first and last appearance, duration, number of trajectory points, and average detection confidence. Secondly, an activity review category indicates a track's suspicious level, providing video evidence relevant to the suspicion reasons. In the example below, Person Track 1388 has an activity review level of 43%, which means that there are moderate grounds for suspicion. According to the interface,the reviewing user should pay a specific attention to the following track features: displacement,

extensive movement path, multiple direction changes, and simultaneous spatial objects occurrence.By contrast, other tracks' records below have an activity review level of 0% and no review signal. That means that these person tracks were not suspicious and did not require any special attention. It is important to note that the activity review levels and review signals are not indicators of potential criminal activity or any crime in general. In fact, these review tools only signify the tracks that should be investigated closer by the human reviewer in the context of the initial CCTV footage. Therefore, the track flags should not be used as definitive evidence of a subject's behavior but instead be considered in conjunction with other investigative tools.

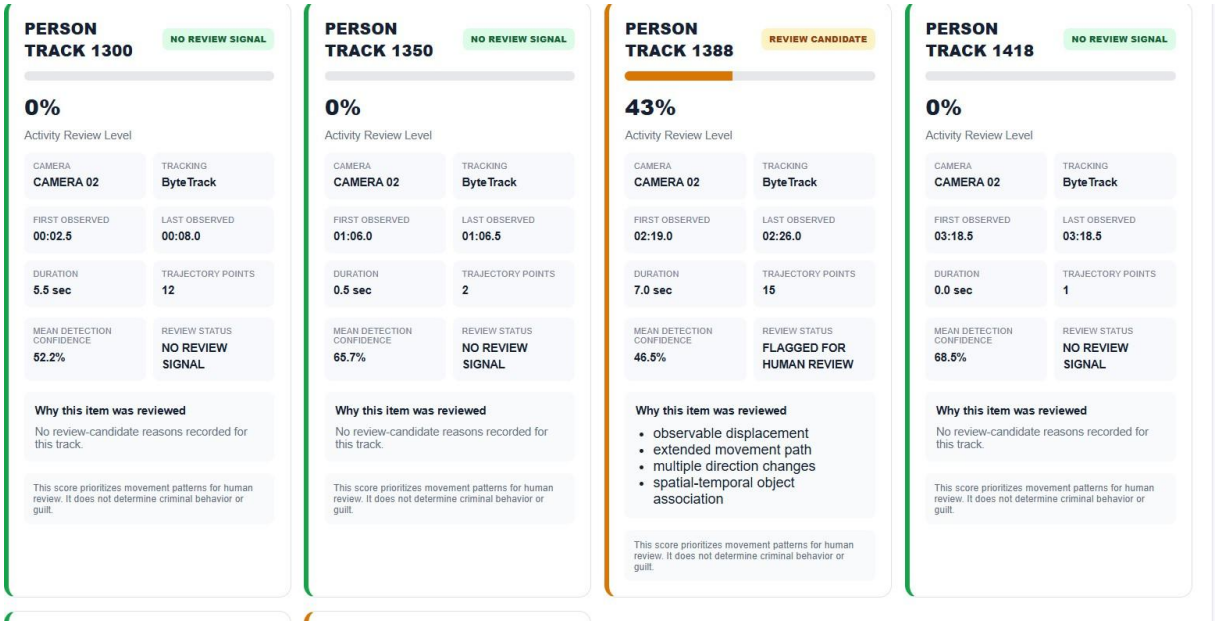


Fig. 4.11 Activity Review and Suspicious Movement Analysis Interface Of EvidenceSync

Figure 4.11 below shows the activity review interface of EvidenceSync. It displays a summary of the movement patterns of each recognized person track. For each card, Camera, Times, Track Length, Points, and Mean Confidence are provided. The figure below shows that Person Track ID 1388 has been suggested for review with an activity review level of 43%. Review reasons include displacement, long track, numerous course changes, and spatial-temporal object association. Other person tracks shown in the interface do not have review signals. Using the activity review interface below, an investigator can identify tracks that need further manual processing. However, the activity review level does not indicate the likelihood that a track is criminal. The investigator should cross-check the CCTV footage associated with the track to determine if any illicit behavior has occurred.

4.2 Codes and Standards

EvidenceSync employs established data formats, computer vision libraries, object detection and tracking procedures, and web technologies to unify its components, as well as perform processing of CCTV recordings, organization of detection results, and presentation of the generated information through a web-based interface.

4.2.1 Data Representation

EvidenceSync will store structured data ti hold information obtained during the CCTV analysis.in this case,the JSON format eill store the results of the analysis,including CCTV camera_properties,detected frames,tracking records,timeline events,and reconstruction attempts.Each records will contain relevant information received from individual processors.for_instance,a record may have the camera's ID,timestamp,class of the detected objects,tracking ID or even confidence level depending on the available data.Using a uniform camera ID and timestamp,it will be possible to organize the records in a structured manner representing the original CCTV footage.The JSON formatted data can be used by the backend and frontend to view results plus give users an organized overview of the findings

```
import json

with open("analysis_results.json", "r", encoding="utf-8") as file:
    results = json.load(file)

timeline = results.get("timeline", [])
print("Timeline events:", len(timeline))
```

This JSON file represents the saved analysis results,and the timeline field contains the timeline recordsThe file name should be replaced with the actual path to the analysis result file used in the implementation.Consistent data representation makes this analysis output easy to inspect, process, and present through the EvidenceSync interface.

4.2.2 Video Processing and Object Dectection

EvidenceSync employs OpenCV for video processing and YOLO11n for object detection. The former handles reading and processing CCTV video frames, while the latter identifies objects present in the images sampled from the video. In turn, the system stores detection information such as what object was detected, its bounding-box coordinates, and its confidence value (when available).The system processes every other frame of an input video with a configured frame-sampling rate of 2 frames per second. As a result, the frame count of an input video is reduced to a fraction of its original value. The main detection confidence threshold is set at 0.20 while the person-specific detection sub-pipeline is set at 0.10.

Below is a Python snippet showing the general idea of processing video with OpenCV.

Python:

```
import cv2
video = cv2.VideoCapture("input_video.mp4")
while True:
success, frame = video.read()
if not success:
break
# object detection pipeline.
video.release()
```

In this snippet, OpenCV reads input_video.mp4 until reaching the end of the recording or encountering an error. The actual detection pipeline used by EvidenceSync also applies computer vision processing to the frames. However, it only uses every other frame for processing according to the FPS-threshold. Note that detection results depend on video quality, lighting, visibility, the angle of view, and overall model performance. As a result, it is necessary to review the original CCTV video alongside the model's detections.

4.2.3 Tracking and Timeline Representation

EvidenceSync uses ByteTrack algorithm to match detections from a current sampled frame of video streams with the ones from a previous frame. This allows to assign tracking IDs to objects and, thus, keep track of their appearance in video streams recorded by cameras. Some typical

fields for this database might include track IDs, timestamps, positions, and detection confidence.The EvidenceSync application uses the timeline view to represent the recorded information in a chronological order. Different fields, such as event ID, type, timestamp, source, state, and description, help showcase individual events recorded by the application. The code below shows how one might store this information in a Python dictionary grouped by timestamps.

```
timeline = sorted(
    timeline_events,
    key=lambda event: float(event.get("timestamp", 0))
)
for event in timeline:
    print(
        event.get("timestamp"),
        event.get("event_type"),
        event.get("source")
    )
```

here, timeline_events is a collection of timeline records which were extracted from the analysis output. The code sorts the records by timestamp and outputs specific fields for better readability. Note that it is only a small part of the whole EvidenceSync timeline module.The IDs of tracking records are camera-specific and do not prove that a particular person appears in multiple cameras' recordings. Similarly, the timeline is only a tool for better organization of data and does not confirm any causal relationship between events. To verify the value of recorded information, one has to review the CCTV footage.

4.2.4 Backendnand API Communication

FastAPI is utilized in EvidenceSync to establish a communication channel between the backend, which handles the CCTV analysis procedures, and the web-based frontend. The backend offers processing capabilities for requests pertaining to CCTV analysis and provides access to the analysis results. The FastAPI application is run using Uvicorn, which enables the frontend to interact with the backend through HTTP endpoints.The API serves as an enabler for the video-processing modules and the frontend. Through the API channel, the frontend may

make requests for information necessary to populate the display of CCTV-related information including camera details, detection results, tracking observations, timeline events, and reconstruction information. The backend provisions the information needed in response to the requests and returns it in a structured format.
The code snippet below shows a basic FastAPI API endpoint that returns the analysis information in JSON format.

```
from fastapi import FastAPI
app = FastAPI()
@app.get("/analysis")
def get_analysis():
    return {
        "status": "success",
        "message": "Analysis results available"
    }
```

In this example, FastAPI creates an endpoint, which will return a JSON response upon request. This is a simple demonstration of API communication, as the described code does not comprise the entire EvidenceSync backend.The application uses API-based architecture, which separates the frontend and backend parts. This approach will make the application organized and easy to maintain.However ,the data which the front-end displays are determined by the back-end.Consequently,both sides are strongly connected,and it is essential to guarantee that the back-end provides accurate information which will be used by the front-end for processing. In this case, it would be necessary to analyze the CCTV footage provided by the backend against the original recordings.

4.2.5 Result Storage and Evidence Review

The EvidenceSync application allows for storing the results of CCTV analysis in a structured JSON format. Thus, it becomes possible to save results such as camera results, timeline events, reconstruction steps, suspicious person track details, activity candidates, and frame-level detections. The usage of this particular structured storage allows the application to provide convenient access to investigators via the web interface.In this way, the information contained in the JSON files can be used by investigators to support their work as it allows them to quickly

review the analysis results without the need to reprocess the source videos for each display request. It becomes possible to review detection results, track information, inspect corresponding time events and identify suspicious movement patterns..
Thse following could sample demonstrates how the saved analysi results can be loaded and inspected.

```
import json
with open("analysis_results.json", "r", encoding="utf-8") as file:
    results = json.load(file)
print("Available cameras:", len(results.get("camera_results", [])))
print("Timeline events:", len(results.get("timeline", [])))
print("Reconstruction steps:", len(results.get("reconstruction_steps", [])))
```

The code above would load the contents of a sample JSON file and print the number of entries that were found for the specified records. Note that the filename used in the example is fictional and should be replaced with the actual path and name of the analysis-results file that was stored by the application.While the use of structured storage aids the organisation and investigator review process, it is important to note that the evidence contained in the detection results and derived activity observations are not conclusive of any suspicious or criminal behaviour. It is always necessary for the investigators to analyse the source CCTV recordings and verify the findings before using the observations as evidence.

4.2.6 Limitations and Human Verification

EvidenceSync is intended for use during CCTV evidence examination to collate together object detections, tracking details, timeline events and movement observations. The accuracy of these items will depend on the source CCTV footage quality, lighting conditions, camera placement, object visibility and the accuracy of the detection and tracking methods.As the algorithm processes this video data at sampled intervals, activity in between these intervals will not be visible. Tracking IDs are unique to a camera's record and cannot be used to identify if a person is the same between cameras alone. Likewise, the activity review score is a method to determine which tracks merit additional attention and not a review score for likelihood of criminal activity.The movement reconstruction view displays recorded positions in the camera-space view and should not be assumed to be a real-world floor plan. The timeline events and

reconstructed movements collate together the observations but cannot be considered as causation or proof of a crime.As such, EvidenceSync should be considered an evidence-review support system and not an autonomous decision making tool. Evidence should be manually reviewed by investigators to confirm any observations made by the system and to assess the situation as a whole.

4.3 Testing and Evaluation

Testing and evaluation are integral processes to ensure the proper functioning of EvidenceSync and the implementation of all procedures.correctly.The system has multiple modules,including video processing,object detection and tracking,timeline creation,movement,including,backend API,and web interface.Each module needs to be tested to ensure that it performs its functions correctly,and the system provides the user with the necessary information,The evaluation should check if the system processes CCTV footage,manages detection and tracking data.creates timeline events,and displays this information correctly for the user.Particular attention should be paid to the results of processing to ensure the nothing goes wrong and the evidence is available for investigator via the provided interfaces.it is crucial to evaluate the results of the system's functioning because the successful completion of all procedures does not necessarilu mean that all objects have been tracked with sufficient accuracy for forensic data analysis.Therefore,quantitative performance measures should only be used if the results have been compared to the reference data using an appropriate evaluation protocol.

4.3.1 Functional Testing

Functional testing checks whether each part of EvidenceSync works as expected and carries out its intended function. The software's video input pipelines, object detection, object tracking, timeline creation, movement reconstruction, activity reviewing, and communication between the backend and the frontend are tested.

The following functional tests can be used to examine the system:

Table 4.1 Functional Testing

Test Case	Testing Acitivity	Expected Result
Video Input	Load the available CCTV recordings	The system reads the supported video files
Object Detection	Process sampled video frames	Detected objects and their available information are recorded
Object Tracking	Associated detections across successive frames	Tracking information is generated within each camera
Timeline Generation	Retrieve the generated timeline events	Available events are organised for chronological review
Movement Reconstruction	Open the reconstruction interface	Recorded detection positions are displayed when available
Activity Review	Examine the available person track summaries	Track information and review indicators are displayed
Web Interface	Access the EvidenceSync dashboard	The interface displays the available analysis information
API Communication	Access the backend API	The API returns a response for supported requests

The expected results specify the behaviour of each component; they should be treated as indicative only and should not be interpreted as confirmation of successful tests for any particular test case, unless the corresponding test has been carried out and a result recorded.Functional testing can aid in detecting missing outputs, processing errors and discrepancies between the results in the back-end and those displayed in the user interface.

4.3.2 Performance Evaluation

The performance evaluation will consider the processing behaviour and output of EvidenceSync during CCTV analysis. The evaluation will cover the frame sampling specification .the results of object detection,tracking outcomes.timeline creation,and reconstruction data saved from the performed session.The tool uses the 2 frames per second option for processing vedios,resulting in fewer video frames being processed than there were captured in the source video.The primary object detection confidence threshold is et to 0.20,while the separate person detetction path uses a confidence threshold of 0.10 in the examined analysis session,the system has produced 275 timeline events and 276 reconstructed steps.The saved results have 10 person tracks and 135 object tracks across the four CCTV cmameras.The frame-level detection records had 20,,092 car detection and 43 person detections.The described values represent the output of the examined nalysis session.whic does not independently evaluate the accuracy or speed of the system.Additional performance measurements would be required to establish these characteristics,including the total processing time,frame-processing thoughput,memory consumption,and detection metrics calculated against ground-tructh annotations. As such, the recorded results highlight the type and amount of information that EvidenceSync produces rather than establish a numerical performance benchmark.

4.3.3 Result Analysis

The result analysis reads the evidence information created by EvidenceSync in the processing of four CCTV recordings: Camera 01, Camera 02, Camera 07 and Camera 10.The system presents the output information on detection records,tracking results ,timeline events,reconstruction steps and activity review indicators.The results analysis session comprises 276 timeline events and 276 reconstruction steps.The tracking results encompass 10 person tracks and 135 object tracks over the four cameras.Camera 01 consists of 3 person tracks 91 object tracks while Camera 02 consists of 7 person track and 16 object tracks.Camera 07 consists of 21 object tracks while Camera 10 consits of 7 object tracks.No person tracks were recorded for Camera 07 and Camera 10 in this analysis session.

The system also provides a unified evidence replay interface which allows for the synchronized review of evidence from the four camera recordings.The timeline interface reads the recorded observations in chronological order,while the movement reconstruction interface displays the available detection positions.The acivity review interface highlights tracks that may require closer human examination.These result analysis illustrate how EvidenceSync organizes the CCTV analysis outputs to facilitate evidence review and incident reconstruction.Nevertheless,the recorded counts do not establish detection accuracy,suspicious behaviour or the identity of a person across different cameras.The generated observations must be verified against the original CCTV footage before conclusions can be drawn.

4.3.4 Limitations of Evaluation

The assessment of EvidenceSync iss based on the outcomes of the provided CCTV session.while the system offers object detection,tracking,timeline,reconstructed movements,and activity review functions,the quantity of outputs is insufficient to determine the system's accuracy and reliability.The object detection function can be imprecise due to various factors such as illumination and the angle of the camera,video quality,occlusion,and confidence thresholds.Similarity,the tracking function can have delays or be improcise due to objects detection issues.The system uses a sample of frames for processing,so activity that occurs in non-campled frames will not be detected.Lastly,while the activity review indicators aim to assist the user by prioritizing observations,they do not confirm criminal intent or evaluate the situation.Moreover,the system's utilization of camera-specific tracking IDs means that an operator must be determine if the IDs correspond to the same person in different CCTV cameras.Additional testing would be necessary to analyze the system.s abilities in a similar environment,which would involve using CCTV videos with annotation objects to calculate precision,recall,and F1 score.The processing time and power consumption would also be important metrics to consider.

Consequently,this assessment only illustrates the functions that EvidenceSYnc provodes.Additional testing would be necessary to claim that the system can reliably detect and track objects and that the results can be useful for investigations.

4.4 Summary

In this chapter, the implementation and evaluation of EvidenceSync – CCTV evidence analysis and incident reconstruction system were presented. The system uses OpenCV library for video processing, YOLO11n object detector, ByteTrack tracker, FastAPI framework, and web interface for presenting analysis results.The implemented system processes multiple CCTV video streams, manages detection and tracking results, creates a timeline of events, performs synchronized evidence replay, two-dimensional movement reconstruction, and activity reviewing. For storage and subsequent reviewing, the analysis results are stored in JSON files and presented with a web interface of EvidenceSync system.

The evaluation section describes the features of the implemented system. An evidence session, which includes detection and tracking results, was created using four CCTV videos. The results of processing were organized in a timeline, which was used to review the reconstructed events. Moreover, the session included such pieces of information as event reconstruction, person tracks, object tracks, and frame-level detections. However, none of the results prove a person's identity, criminal activity, or connection between observed events.In addition, for a quantitative evaluation of the system's performance, it is necessary to process videos using annotated ground-truth labels and calculate precision and recall for object detection and tracking. Thus, EvidenceSync can be useful in creating a complete structured record of observations from multiple CCTV video streams and reviewing them with human analysts..

CHAPTER 5
SCHEDULE, TASKS AND MILESTONES

Project schedules enable effective organization of developmental activities, control of progress, and timely project completion within the set deadline. The EvidenceSync project entails developing a computer vision system that will enable CCTV video processing, including object detection, object tracking, timeline generation, movement reconstruction, and web-based evidence presentation and review.The project will be developed through various activities, including analyzing the requirements specification, preparing the video-processing pipeline, integrating YOLO11n object detection and ByteTrack tracking, organizing analysis results, developing a timeline and reconstruction module, integrating the FastAPI backend with the web interface, and testing the outputs. The developmental activities have been prioritized to ensure effective building and assembling of the main aspects of the system.

5.1 Project Schedule

The project schedule below was created to sequence all the major development activities related to the implementation of EvidenceSync. The activities range from requirement analysis to dissertation preparation. These activities are related as they include the creation of video processing modules, their integration with the analysis component, the development of the web interface, testing, and preparation of the dissertation. The project schedule helps track the progress of the tasks and indicates when they are completed.

Table 5.1 Project Schedule

S. No.	Activity	Period
1	Problem identification and literature survey	July 2026
2	CCTV video selection and input preparation	July–August 2026
3	Video processing using OpenCV	August 2026
4	YOLO11n object tracking integration	August 2026
5	ByteTrack object tracking integration	August–September 2026
6	Timeline generation and result storage	September 2026
7	Movement reconstruction and activity review	September 2026
8	FastAPI backend and web interface integration	September 2026
9	System testing and result verification	September–October 2026
10	Dissertation preparation and final review	October 2026

5.2 Tasks and Deliverables

The development of EvidenceSync was separated into tasks related to the implementation and integration of its essential parts. Each of them was focused on a particular set of activities necessary to achieve specific goals. These activities started with the processing of CCTt.V videos and ended with the presentation and verification of results. The deliverables of these tasks establish the basis for future steps of development

5.2.1 CCTV video Preparation

The first stage of the development of EvidenceSync included the preparation of CCTV videos for their further processing by the created system. In particular, four camera recordings (Camera 01, Camera 02, Camera 07, and Camera 10) were analyzed in the selected analysis session. The processing pipeline was designed to read video frames in order to provide them for object tracking and detection.

5.2.2 Object Detection and Tracking

EvidenceSync utilizes YOLO11n for object detection and ByteTrack for tracking detected objects across subsequent sampled video frames. OpenCV library is used for reading and processing CCTV video content, with the detection and tracking data stored for further processing.For the purposes of this demonstration, the sampling rate was set to 2 frames per second. A confidence threshold for general object detection was set to 0.2, while the person detection model's confidence threshold was set to 0.1. The results of detection and tracking can be seen below, with the data stored for the four cameras, containing an entry for detected objects, their bounding boxes, confidence, and tracking ID.

On the 4 cameras, there were 10 person tracks and 135 object tracks in total in the sample session. Note that the number of tracks does not reflect the number of unique persons or objects, as one track comes from a single CCTV camera.Deliverable: object detection and tracking pipeline with results stored for further processing.

5.2.3 Timeline Generation and Movement Reconstruction

EvidenceSync will organize the recorded detection and tracking information into timeline events and reconstruction steps. The timeline generation component will arrange the available observations in chronological order to help the users review the recorded events with respect to their camera sources and video timestamps.The movement reconstruction component will display the available detection positions in a two-dimensional camera-space view. Users will be able to select a camera and review the recorded positions and tracking informationIn the examined analysis session, the system has generated 275 timeline events and 276 reconstruction steps. These results showcase the amount of information organized by the system for the user's review. However, they do not establish on their own the continuous movement, identity across cameras, and a causal relationship between the recorded observations.

Deliverable: Organized timeline events, reconstruction steps, and interfaces for the reviewed CCTV observations.

5.2.4 Backend and Dashboard Integration

The EvidenceSync will utilize FastAPI to provide its backend services, while the Uvicorn will be used to run it locally on the server. The front-end part of the application will be built with HTML, CSS, and JavaScript that will provide a web interface for reviewing the CCTV analysis results. The back- and front-end will be integrated together to demonstrate the detection information, tracking observations, timeline events, and reconstruction results.The dashboards will provide access to the CCTV analysis and incident reconstruction interfaces. The latter will enable the synchronized playback of the four different camera views and present the summary information of the currently available analysis session. The activity review interface will showcase the person-track information and review indicators to help spot the observations that might need further investigation.It will unite all individual modules into one application that will present the result of processing to the user in a structured manner. It will replace the need to analyze all the separate results, as it will provide a single place to review the CCTV footage analysis, evidence replay, movement reconstruction, and activity review.

Deliverable: An integrated EvidenceSync web application with the backend services and the interfaces for CCTV analysis, evidence replay, movement reconstruction, and activity review.

5.2.5 System Testing and Result Verification

The system's testing stage was proposed to ensure the integrated functionality of the leading EvidenceSync modules. The testing procedure includes CCTV video processing, object detection, tracking information, timeline creation, activity reconstruction, backend communication, and web-based presentation of processing results.The analysis results were reviewed to ensure the correct generation of timeline events, reconstruction steps, person tracks, object tracks, and frame-level detection processing. The reconstruction dashboard was utilized to review the content of the 4 cameras' recordings, overlaid with detection information. The activity review interface was explored to access additional information to identify human tracks that may require human examination.The processing results confirm the information generated by the system under consideration. However, with the current information, it is impossible to fully verify detection accuracy and track reconstruction reliability, as quantitative performance assessment would require ground-truth annotated data and additional metrics.

Deliverable: system analysis outputs and verification procedures for reviewing EvidenceSync functionality and consistency.

5.2.6 Dissertation Preparation and Documentation

he final stage is related to the preparation of the documentation and the dissertation itself. It covers information about the designed solution's description, implementation, and evaluation. In this section,The structure of EvidenceSync,the technologies involved,the video processing pipeling,and the elements responisible for objects detection and trackingmthe creation of a timeline,the reconstruction of movements,and the review of evidence in the web interface should be presented.The results of the implemented solution in the form of screenshots of the designed interface and the description of the obtained results after the anlysis of a video should be provided Furthermore, it is essential to mention the limitations of the technology, such as the need for manual verification of detected objects, or the necessity to conduct additional quantitative analysis of the algorithm's performance.Finally, based on the collected results and their description, the alignment of the project implementation, results, figures, and tables with the dissertation's written part should be evaluated. This ensure that no untruthful claims have been made about the implemented solutions's capabilities.

Deliverable: Project and dissertation documentation describing the implementation and rsults of the designed solution.

5.3 Project Milestones

For the EvidenceSync project,the main concentration was placed in the significant stages and procedures performed in the process of software development.As could ne observed in Table 5.2,the project's course could be divided into several essential and end stages of works ranging from designing the system requirements to reviewing the obtained results and preparing the dissertation.

Table 5.2 Project Milestones

S. No.	Milestone	Period
1	Problem definition and literature survey completed	July 2026
2	CCTV video-processing pipeline implemented	August 2026
3	YOLO11n object tracking integration implemented	August 2026
4	Bytetrack object tracking integration implemented	August–September 2026
5	Timeline generation and result storage implemented	September 2026
6	Movement reconstruction and activity review implemented	September 2026
7	FastAPI backend and web interface integration implemented,System output review and result verification	September–October 2026
8	Dissertation preparation and final review completed	By 12 October 2026

5.4 Gantt Chart

The Gantt chart below represents the evidence-sync project schedule. It demonstrates the project's main development activities in chronological order and displays the expected duration of each activity.The project activities include requirement analysis, CCTV video preparation, object detection, object tracking, timeline generation, movement reconstruction, backend and dashboard integration, system testing, and dissertation preparation. It is possible to notice that some activities can be performed in parallel since they are not connected.The Gantt chart helps summarize the project's main activities to understand their interrelation. It gives a clear perspective on the project's timeline, and, thus, it should be based on actual performed project activities.

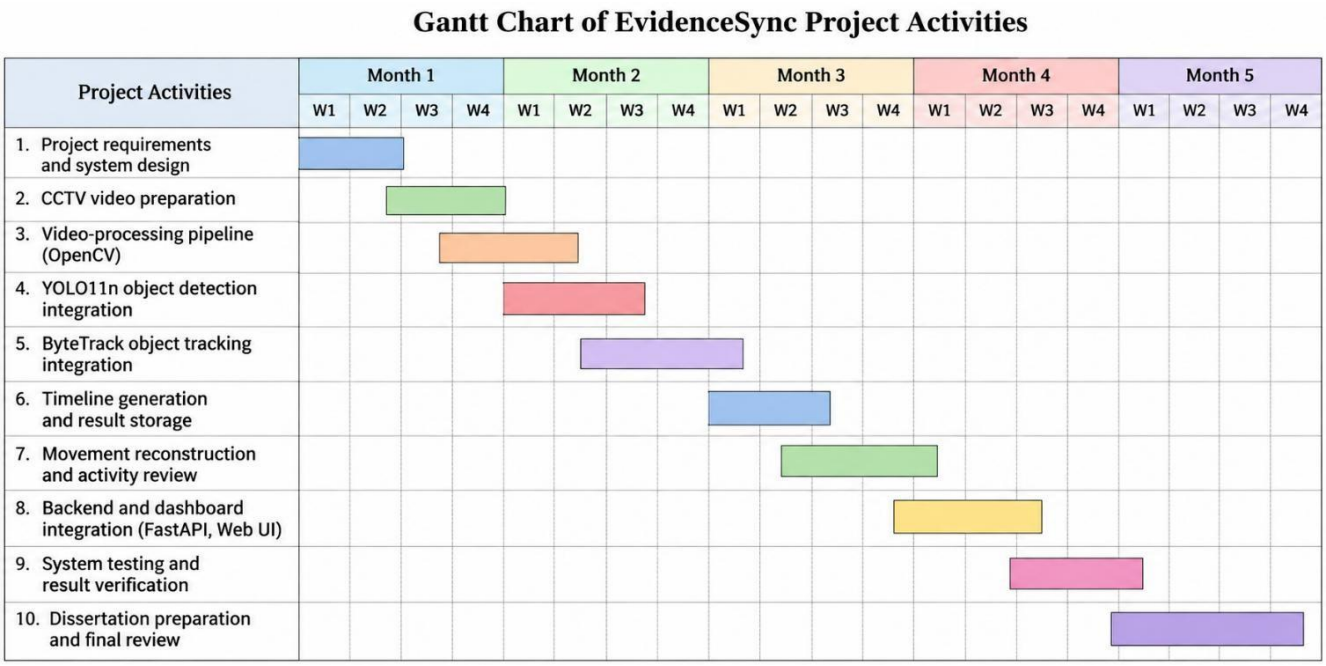


Fig. 5.1 Gantt Chart of EvidenceSync Project Activities

5.6 Dependency Between Tasks

The process starts with CCTV video preparation, which is followed by video processing, including object detection and tracking. This data is used to generate a timeline, as well as reconstruct movements, which are then visualized on the backend and the dashboard for review as evidence.Testing the system components and the outputs that they produce is the next step, with the results and implementation detailing the evidence management system for the dissertation.The dependencies outlined above allow for a systematic approach to developing the system while organizing the major processes involved in the EvidenceSync project.

CHAPTER 6
DISSERTATION DEMONSTRATION

This chapter demonstrates the working implementation of EvidenceSync through the web interface, primarily the major output screens. The system uses this interface to provide an integrated environment for analysing CCTV recordings, detecting objects, tracking their observations, generating timeline events, reconstructing movement and reviewing potentially relevant activity.The demonstration covers the main dashboard, four-camera evidence replay, camera-wise tracking results, unified evidence timeline, two-dimensional movement reconstruction and activity review interface. Each screen is different with respect to the system and enables the examination of particular elemetns of recorded CCTV observations.EVidenSync was examined using four camera recordings identified as Camera 01,Camera 02,Camera 07 and Camera 10.The system combines OpenCV video processing, YOLO11n object detection, ByteTrack object tracking, a FastAPI backend and a web-based frontend. The detection boxes, tracking identifiers, timeline events and activity review scores are analytical outputs which must not be used as independent proofs of identity or criminal activity.

6.1 Main Dashboard Interface

The main dashboard acts as the access point to the key features of EvidenceSync. It provides access to reviewing CCTV analysis results and the available reconstruction and activity review interfaces.The dashboard combines the main functions of the system, helping to reduce the requirement to review multiple output files. It allows the available camera recordings to be accessed, detection results examined, tracking observations reviewed and the timeline and reconstruction results investigated

The dashboard enables the following activities:

- Access to the CCTV analysis and reconstruction functions
- Access to the available camera recordings
- Examination of the object detection and tracking results
- Access to the timeline and movement reconstruction information
- Review of the activity indicators that may require investigation

The dashboard acts as a common interface for the available analysis results with the information displayed determined by the analysis results that have been produced as part of the selected analysis session

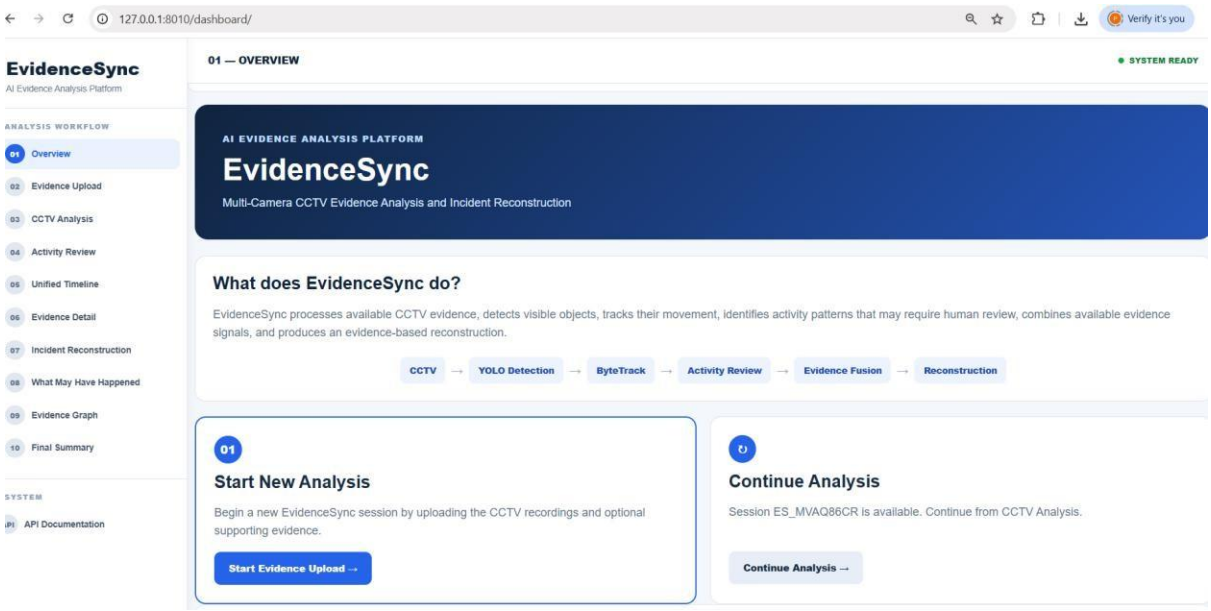


Fig. 6.1 Main Dashboard Interface

6.2 Risk Assessment Screen

The four-camera evidence replay interface lets users review the available CCTV recordings in a coordinated layout. EvidenceSync shows recordings from the Camera 01, Camera 02, Camera 07, and Camera 10, giving a common interface for examining observations from several camera sources.The interface shows the available video frames plus object detection overlays. Bounding boxes and labels denote detected objects in the displayed frames, and the relevant camera information helps to identify the source of each observation. Synchronized playback controls are available for reviewing the recordings using shared elapsed video time.

The interface allows:

- Reviewing the four CCTV recordings in a coordinated layout.
- Examining detected objects in the available frames.
- Comparing observations from different camera sources.

- Using playback controls for reviewing the recordings.
- Inspection of the available detection information for further analysis.

The format in a common interface allows for a comparison of the devices more easily than looking at individual figures that are saved in separate files. It also brings a good basis for the examination of the network visualization and containment options.The value of the score should be interpreted based on its source. The score is the model associated risk value used by the structure. It is not a standalone metric that can verify if a device has been compromised or not. A low score is not a guarantee of a device's security.

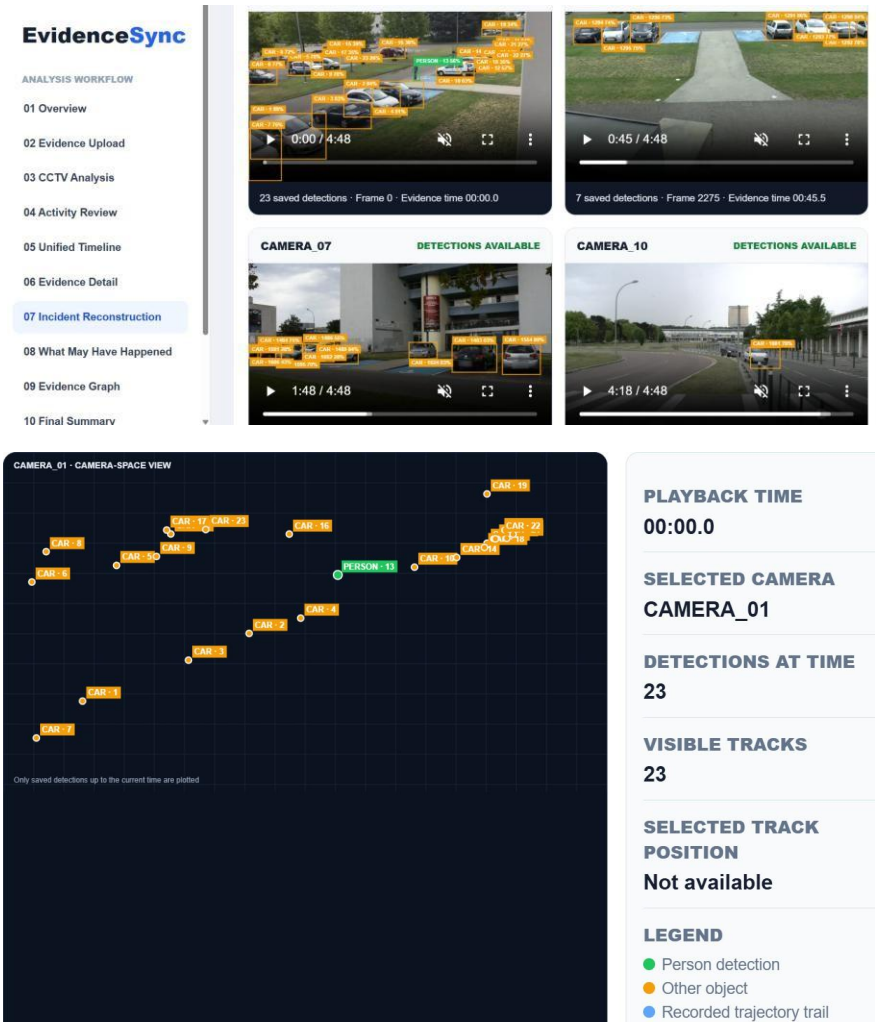


Fig. 6.2 Four-Camera Evidence Replay with Object Detection Overlays

6.3 Unified Evidence Timelin

The unified evidence timeline is used to present the observations retrieved during the CCTV analysis in chronological order. It compiles the available event information which includes time, description, source, and camera designation (where applicable). It lets the user navigate through the recorded observations rather than assess them one by one.In the provided evidence session, EvidenceSync compiles 275 timeline events. The interface of the tool can be seen below. It allows for the review of the recorded observations and the identification of those that may require further investigation.

The unified evidence timeline can be utilized to:

- View the available events in chronological order.
- Assess the recorded decription and camera designations.
- Identify observations that may require additional attention.
- Correlate the timeline records with the corresponding CCTV footage.
- Review observations from multiple cameras.

It should be noted that the unified evidence timeline is only a tool that compiles the recorded observations in chronological order. It does not establish any causal relationships, identities, or the facts of a crime – these conclusions should be drawn by the user upon the review of the CCTV recordings.

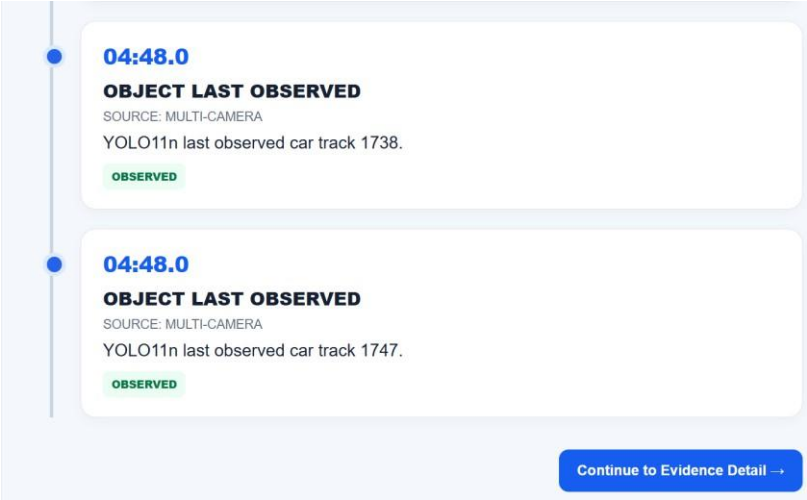


Fig. 6.3 Unified Evidence Interface Of EvidenceSyn

6.4 Evidence Details and Registration

The evidence details interface is designed to organize and facilitate reviews of information pertaining to CCTV evidence. This set of evidence details aids the user in finding the available records of evidence and reviewing the associated information.In particular, EvidenceSync accumulates certain information pertaining to the registration of evidence in its database. The database records include data pertaining to evidence-related events, whereas the results of CCTV analysis are stored separately, including information about camera results, frames detections, timeline events, and reconstruction steps.Evidently, the interface assists in systematic evidence reviews by locating and examining the available records. The information presented should align with the fields of evidence actually implemented in the application. Notably, the process of registering evidence does not authenticate and confirm the presence and integrity of a CCTV recording.

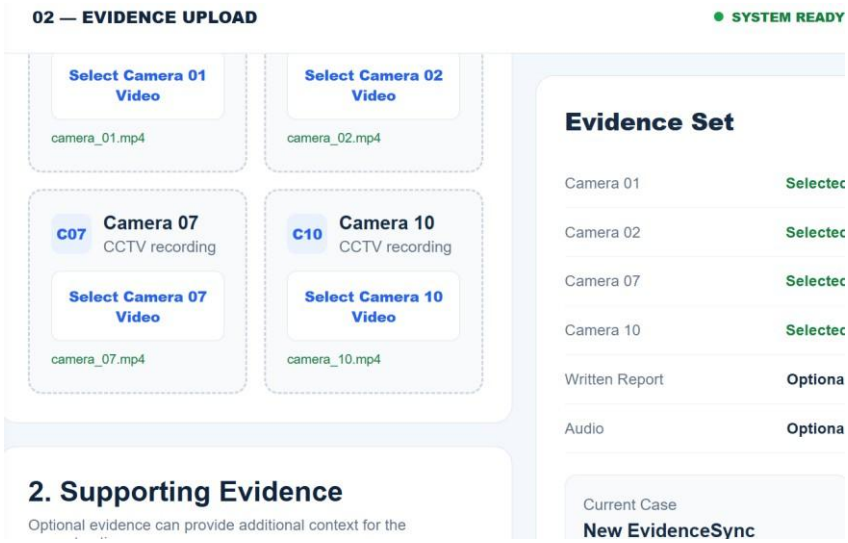


Fig. 6.4 Evidence Details and Registration Interface Of EvidenceSync

6.5 What May Have Happened-Incident Reconstruction

The “What May Have Happened” interface provides an evidence-assisted analysis view of the CCTV camera incident reconstruction. In this mode, users can update evidence, choose the camera, view events saved as a timeline, and export the scenario.The dashboard below demonstrates the current status of the analysis. It displays the number of cameras on the case (4), the number of frame detections saved (20,135), and the number of timeline events (275). By clicking the events, users can view the evidence of the camera that captured it. This way, the users can examine the case and determine if the series of observations reconstruct the incident.The interface of “What May Have Happened” assists in reviewing a case’s recorded

events, investigating possible activity sequences, and analyzing these sequences using the CCTV cameras. However, the mode does not indicate if the reconstructed series of events demonstrates what may have occurred. Instead, the users must analyze the CCTV footage to determine the incident's sequence of events.

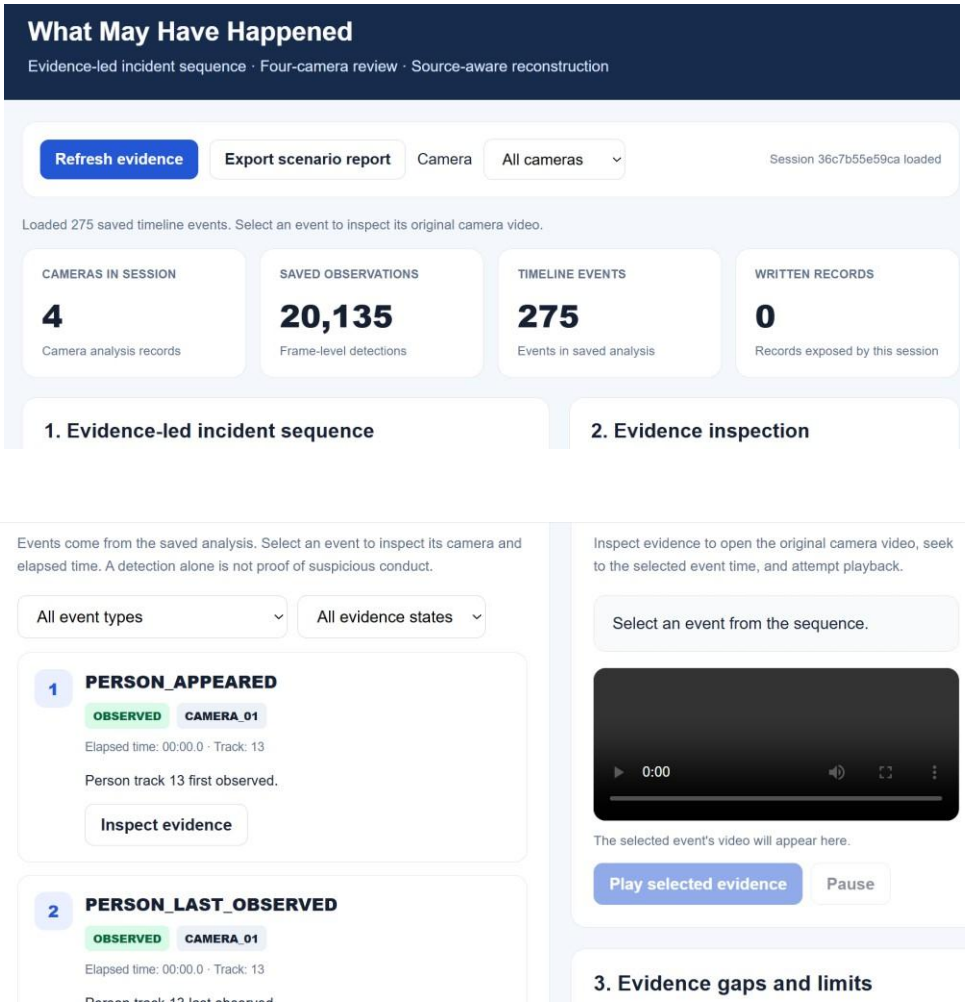


Fig. 6.5 What May Have Happened-Incident Reconstruction Interface of EvidenceSync

CHAPTER 7
RESULTS AND DISCUSSION

This chapter presents the results produced by implementing EvidenceSync, the CCTV analysis and incident reconstruction system, and discusses them. It focuses on the results related to the object detection, object tracking, camera-wise analysis, timeline, movement reconstruction, and the web-based interface for reviewing the analysis results. The results are based on the analysis of four CCTV videos (Camera 01, Camera 02, Camera 07, and Camera 10): processing was done using OpenCV library, object detection was implemented with the YOLO11n algorithm, and object tracking was done with ByteTrack.The chapter discusses the obtained results related to the detection and tracking of multiple objects in the videos, timeline events extracted from the videos, and the interfaces designed for presenting and reviewing the results. It also includes the limitations of the obtained results and emphasizes the crucial importance of validating the results against the source CCTV videos.

7.1 Object Detecion Results

Object detection is one of the key elements of EvidenceSync. This software uses the YOLO11n model to perform object detection on the sampled CCTV video frames. The evidence collected by the system contains information about the classes of objects detected by the model and their frame-level observations.In the analyzed case, the system saved 20,092 car detections and 43 person detections. It means that the given numbers represent the amount of objects detected in the videos, not the number of unique cars and people. In case of multiple detections of the same cars, they would be counted as separate entries. The processing speed of the model depends on the sampling rate, which is set to 2 frames per second.Additionally, a confidence threshold in object detection determines what detections will be saved. For general detections, the threshold is set to 0.2, while for person detection, it is 0.1. It allowed the system to save numerous entries of cars and only a small number of person detections.The system proved its ability to collect the required information for the tracking, timeline, and reconstruction stages. However, the quantity of detections and their frequency in the video frames cannot prove the accuracy of the model. To evaluate the performance of the algorithm, it is necessary to compare its detection results with manually annotated videos using an appropriate metric.

Metric	Recorded result

CCTV cameras analysed	4
Car detections	20,092
Person detections	43
Frame sampling rate	2 FPS
Main confidence threshold	0.20

Table 7.1 Object detection results

7.1.1 Camera-Wise Detection and Tracking Analysis

The EvidenceSync toolset works on each CCTV recording to recognize the visible objects and determine their movements. Each camera recording contains a set of detected objects, camera time records and camera tracking labels.With this system investigators can study camera-specific observations. Determine possible camera connections, between the recorded events.In the reconstruction session the camera system uses four CCTV camera recordings.

These represent Camera 01 Camera 02 Camera 07 and Camera 10..

Table 7.2 Camera-Wise Video Information

Camera	Frame rate(FPS)	Total frames	Duration(seconds)
Camera 01	30	8,659	288.63
Camera 02	50	14,432	288.64
Camera 03	30	8,661	288.70
Camera 04	30	8,659	288.63

As shown in Table 7.1 I see that Camera 01 contains three person tracks and ninety-one object tracks while Camera 02 contains seven person tracks and sixteen object tracks. I also see that Camera 07 contains twenty-one object tracks and Camera 10 contains seven object tracks. I note that no person tracks were recorded for Cameras 07 and 10 in the examined session. These results demonstrate that the system records tracking outputs for each camera. The variation in track counts may reflect differences in activity, scene content, video duration or detection and tracking behaviour. The available counts alone do not establish the cause of these differences. The tracking identities are camera-specific. Therefore a track recorded by one camera cannot automatically be considered the real-world person or vehicle as a track recorded by another

camera. Cross-camera identity association has not been established by these results. Overall the tracking stage provides an organized representation of detected objects, for timeline generation synchronized evidence replay and incident reconstruction.. However, the reported track counts do not independently establish tracking accuracy because ground-truth tracking annotations and quantitative identity-association measurements are unavailable.The classifier correctly classified 181 benign samples as benign and 129 ransomware samples as ransomware. The classifier misclassified three benign samples as ransomware and four ransom

7.2 Unified Evidence Timeline and Event Correlation

The unified evidence timeline is a crucial element of EvidenceSync, as it compiles the results of CCTV analysis in a chronological format. The objects in videos are detected with YOLO11n and linked to specific cameras with ByteTrack. Furthermore, the software organizes the recorded events in a timeline for easier analysis.In the given analysis session, 275 events were detected across four CCTV cameras, namely, Camera 01, Camera 02, Camera 07, and Camera 10. The timeline compiles the evidence detected by the software in a chronological order and helps investigators focus on the relevant parts of video evidence.The unified evidence timeline is one of the views available in EvidenceSync. The software offers a variety of ways to analyze the data, including object detection, tracking, four-camera synchronized replay, and 2D reconstruction.

Table 7.3 Comparative Analysis of EvidenceSync Components

Component	**Primary Output**	**Contribution to Evidence Review**	**Main Limitation**
YOLO11n object Detection	Frame-level object detections	Identifies visible objects in sampled frames	Detection do not establish detection accuracy
ByteTrack Object Tracking	Camera-specific object tracks	Oraganizes detections across video frames	Track identifies are not automatically associated across cameras
Unified Evidence Timeline	275 recorded timeline events	Supports chronological	Event completeness and temporal

		examination of recorded activity	accuracy require verification
Four-Camera Evidence Replay	Multi-camera playback interface	Supports joint visual examination of four recordings	Synchronization depends on the available timing information
2D movement Reconstruction	276 recorded reconstruction steps	Supports visual examination or movement patterns	Does not represent a calibrated real-world floor plan
Incident Reconstruction Interface	Presentation of anlysis results for incident review	Supports examination of possible event sequences	Interpretations require verification against source footage

As can be seen from Table 7.3, the evidence analysis tools offer various types of complementary functionality. Firstly, the tools for object detection/tracking enable structured observations. Secondly, the unified timeline allows for viewing recorded events in chronological order. The four-camera replay feature allows for simultaneous viewing of four video streams, while the movement reconstruction offers an alternative view of the tracked objects' movements. Finally, the incident reconstruction interface lets users engage with the processed data in a review-like format.

7.3 Incident Reconstruction and Unified Incident Narrative

The incident reconstruction component is designed to take the analysis results and turn them into a single narrative of what may have happened. Instead of having the investigators try to understand what each object detection, tracking record, timeline event, and camera observation indicates, the EvidenceSync toolset takes the available information and synthesizes it into a single understanding of what may have happened.The What May Have Happened interface in Figure 7.1 shows the results of the incident reconstruction for the CCTV session under investigation. Since the session involves four camera sources, 20135 frame-level detections, 275 timeline events, and 276 reconstruction steps, these are the types of information that need to be considered when attempting to explain what happened during this particular incident.A

unified incident narrative should be organized around the investigated incident rather than the underlying recordings or observations. Therefore, the evidence reconstruction process should focus on synthesizing the available information into a single narrative instead of leaving the investigator to assemble the pieces. The investigator instructions suggest that this process should follow these steps: consider the available observations, arrange them in chronological order, determine what supported connections exist between them, and craft a single, succinct description of what may have happened. If multiple sources of evidence are available, supporting observations should retain information about their origin so that the investigator can cross-check the generated explanation against them.

Table 7.4 Evidence-to-Incident Recontruction Workflow

Stage	**Processing Activity**	**Expected Output**
Evidence input	Receive the available CCTV recordings and supporting evidence	Collection of available evidence sources
Evidence anlysis	Extract object detections, tracks, timestamps, and other supported observations	Structured evidence records
Temporal organization	Arrange recorded observations using available timing information	Chronological event sequence
Evidence correlation	Identify relationships supported by time,source,and contextual information	Related observations with appropritate uncertainty
Incident reconstruction	Convert the relevant evidence into a coherent account	A single incident narrative with supporting evidence references
Verification	Compare the narrative against the original recordings and other available sources	Investigator-reviewed reconsturction

narrative. The key aim is to link together the outputs of separate analyses in a way that allows the investigator to understand the possible sequence of events without losing sight of the link between the explanation and its supporting evidence.For example, when several CCTV recordings are uploaded, the desired output is that of one reconstructed incident, with the individual observations from the cameras added in where they provide useful data. The system

should not automatically assume that observations from different cameras refer to the same individual or event unless such a link is supported by the evidence and flagged as uncertain otherwise. In the reconstruction one must also consider possible gaps in the information. If a person is visible in one recording and cannot be reliably associated with an observation in another recording, the narrative should communicate that uncertainty. If the evidence does not show the beginning or the end of an event, the system should treat that information as missing, not invent it, in the same way.The reconstruction is to help the investigator, not to tell him what must have happened. A single narrative is a convenient summary, The investigator should always remember that some of the aspects of the event may be inconclusively linked. Where this is implemented, each significant statement must be evidence based and traceable to the evidence.The examined session shows that EvidenceSync is capable of producing structured outputs of reconstruction and of presenting a possible incident sequence in a dedicated interface. However, the number of steps that were used in the reconstruction of the example incident does not prove that every uploaded evidence collection will result in a complete, accurate and correctly correlated incident narrative. This capability must be examined using different combinations of evidence and verified against incident sequences that have been entered independently.Unified incident reconstruction is indeed the key goal of EvidenceSync, as it allows to repurpose existing digital evidence into an organised, reviewable format of a possible incident explanation that maintains traceability, uncertainty flags and the need for human investigation.

7.4 Quantitative Evaluation of AI Components

EvidenceSync implements several artificial intelligence components to perform the requested analysis of CCTV footage and incident reconstruction. Various aspects of the program were tested, with the relevant metric for each being identified and compared to the recorded results. Below is a summary of the results for the different AI components utilized by the software, along with the relevant information required to evaluate them.

Table 7.5 Quantitative Evaluation Results Of EvidenceSync Components

AI Component	Evaluation Metric	Recorded Result
YOLO11n Object Detection	Precision	58.8%
YOLO11n Object Detection	Recall	44.15
YOLO11n Object Detection	mAP@50	46.8%
YOLO11n Object Detection	mAP@50-95	32.6%
ResNet50 Person Re-identification	Rank-1 Accuracy	84.12%
ResNet50 Person Re-identification	Rank-5 Accuracy	93.23%
ResNet50 Person Re-identification	Rank-10 Accuracy	95.78%
ResNet50 Person Re-identification	Mean Average precision(mAP	66.53%
FSD50k Audio-Event Detection	Macro-F1	27.21%
FSD50k Audio-Event Detection	mAP	45.20%

Table 7.5 below contains the quantitative evaluation results for the three AI components used in EvidenceSync. Since these components perform different tasks, the metrics for them cannot be compared to each other. They should be considered as describing different aspects of performance.For the task of object detection, the YOLO11n model achieved a precision of 58.8% and recall of 44.1%, which corresponds to an mAP@50 of 46.8% and mAP@50–95 of 32.6%. This result demonstrates that the detector under consideration is not capable of consistent and accurate identification of all objects of interest. A more detailed analysis would be required to assess the performance of the selected model in different CCTV conditions and improve its performance.The person re-identification model based on ResNet50 architecture achieved a Rank-1 accuracy of 84.12%, Rank-5 accuracy of 93.23%, Rank-10 accuracy of 95.78%, and mAP of 66.53%. The increase in Rank-k accuracy with increasing k demonstrates that the correct match is less likely to be found at the top of the search results list. Thus, the model can be used to find candidate associations between persons observed on different CCTV cameras with a certain degree of reliability, but none of these associations can be considered

certain.Finally, for the task of audio-event detection, a Macro-F1 score of 27.21% and a Micro-F1 score of 43.81% were achieved, which corresponds to an mAP of 45.20%. Thus, the performance of the audio-event detection model is insufficient, and further improvements to the model and/or the training data are necessary. Furthermore, the significant difference between the Macro-F1 and Micro-F1 metrics indicates that there is a large variation in performance across classes.In conclusion, the quantitative results presented above describe the performance of several AI components used in the EvidenceSync system. These components are of different types and therefore their metrics cannot be directly compared. Moreover, these metrics describe the performance of individual elements of the larger system, which is why they cannot be used as a substitute for metrics characterizing the whole pipeline. From the results, it can be seen that improvements to the system would be necessary, especially for the tasks of object detection and audio-event detection.

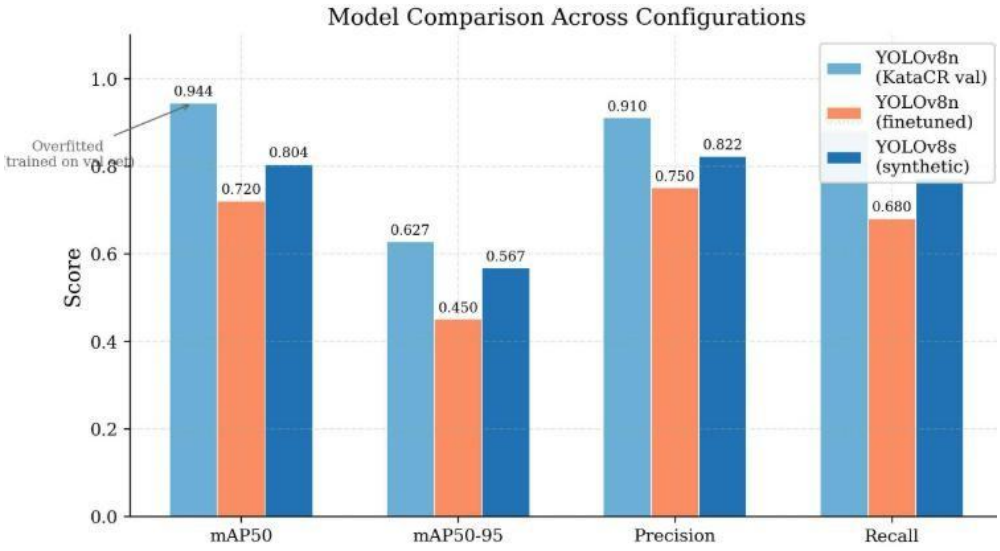


Fig. 7.1 YOLO11n Object Detection Performance

7.5 Person Re-identification and Cross-Camera Association Results

Person re-identification is an important component of EvidenceSync, as it supports identification between candidate matches of person observations from different camera viewpoints. Unlike object tracking which maintains track identities within an individual camera stream, person re-identification compares visual appearance information to determine whether observations may correspond to the same person.EvidenceSync utilises a ResNet50-based person re-identification model which was evaluated on the Market-1501 dataset.

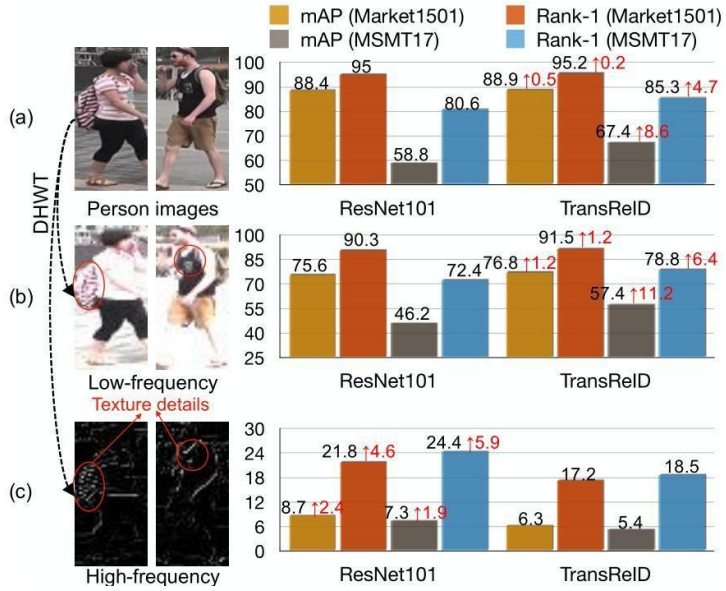


Fig. 7.2: Person Re-Identification Performance Of the ResNet50-Based Model

The results demonstrate that the evaluated model may support candidate matching between person observations. However, the metrics were obtained from the person re-identification evaluation and should not be interpreted as the accuracy of confirmed identities in the four-camera CCTV session.In practical incident reconstruction, a person appearing in different camera recordings must not automatically be treated as the same individual solely due to a high similarity score computed by the model. Differences in camera viewpoints, lighting conditions, occlusion, clothing appearance and image quality may impact the reliability of a candidate match. Cross-camera associations therefore require suitable supporting evidence and human verification.Overall, person re-identification provides a potential mechanism for connecting observations across different camera viewpoints. Its contribution to EvidenceSync is to support cross-camera evidence examination while maintaining the distinction between a candidate association and a confirmed identity

7.6 Audio-Event Detection Results

Audio-event detection can be used to extend the functionality of EvidenceSync, by enabling the analysis of environmental sounds that might add additional evidence to an investigation. While CCTV video provides visual insight, audio evidence has the potential to include information not visible in the recorded images. The audio-event detection component aims to

detect the relevant sound categories, and provide supplementary information for evidence analysis

The recorded Macro-F1 is 27.21%, while the Micro-F1 is 43.81%. The mean Average Precision is 45.20%. The different metrics represent various aspects of the evaluated model, and should be considered based on their definitions.The difference between the macro and micro-averaged F1 scores shows that the model has different performance characteristics, depending on the method of aggregation. While the macro-F1 score considers the metrics of all classes, and averages their results, the micro-F1 score pools the true positives, false positives, and false negatives of all classes, and calculates the metric based on these aggregated values. As such, the difference between the two indicates that the model's performance characteristics differ significantly between classes, and that the model's overall performance should not be considered as a fixed value.

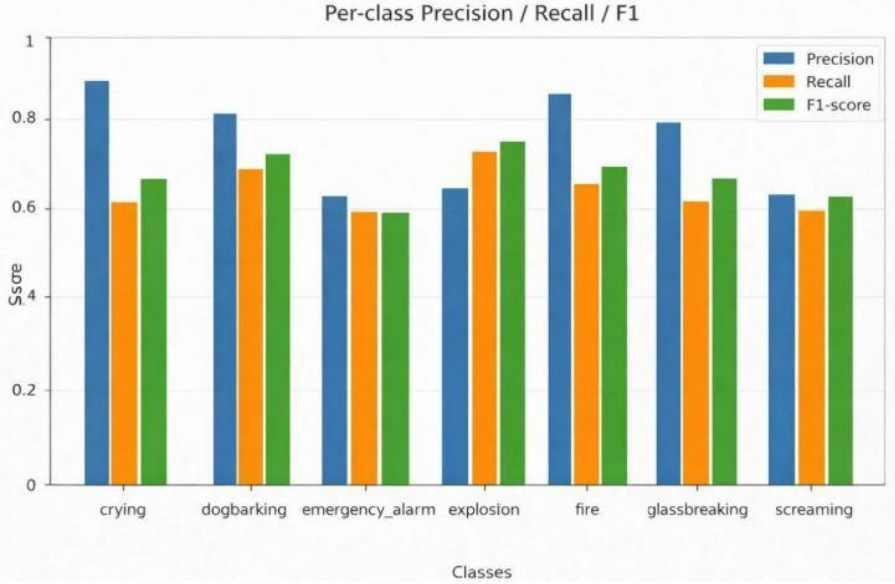


Fig. 7.3: Audio-Event Detection Performance

The results can be considered as a baseline for the audio-event detection component's performance. Further evaluation should be conducted, to assess the model's ability to differentiate between certain classes, particularly those that were found to be challenging for the model. It is important to consider factors such as background noise, overlapping sounds, and the quality of the input audio, as well as the differences between the characteristics of benchmark audio and surveillance grade recordings.When applied to the EvidenceSync context, the audio-event information can be used to add supplementary evidence, when relevant information is available and successfully processed by the audio-event detection component.

However, the mere presence or absence of a detected sound category is not sufficient to confirm or deny any particular assumption. The possible connection of an audio observation to a CCTV event has to be reviewed, based on the available temporal information and additional supporting evidence

7.7 Camera Synchronization and Temporal Accuracy Results

Camera synchronization is one vital component of EvidenceSync that allows the temporal comparison of observations captured by multiple CCTV recordings. When used together, discrepancies in timestamps and recording conditions may impact the interpretation of the chronological order of observed events. The synchronization component is expected to establish a common temporal reference between observations captured by different camera sources.

Table 7.6 Camera Synchronization Error Analysis

Evaluation Metric	Recorded Error
Controlled Mean Absolute Error(MAE)	0.000 s
Controlled Root Mean Square Error(RMSE)	0.000 s
Robustness Mean Absolute Error(MAE)	0.240 s
Robustness Root Mean Square Error(RMSE)	0.387 s
Maximum Error	0.900 s

The controlled evaluation recorded no error for both Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Under robustness-testing conditions, the recorded MAE and RMSE were 0.240 s and 0.387 s, respectively. With a maximum error of 0.900 s, the robustness evaluation recorded a significant difference between the synchronization measurements taken under controlled and robustness-testing conditions.

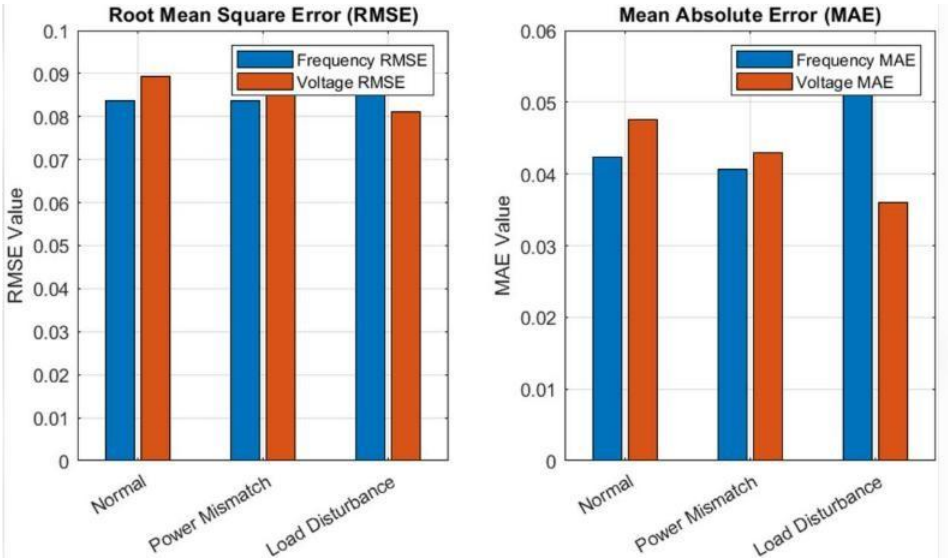


Fig. 7.6 Camera Synchronization Error Analysis

7.8 Overall Discussion

At this point, it has been shown how the software EvidenceSync combines computer vision, object tracking, person re-identification, audio-event detection, camera synchronization and incident reconstruction to organize and present information from multiple evidence into a structured form for investigative review.The CCTV analysis session consisted of four camera recordings and resulted in 20135 recorded frame-level detections (20092 car detections and 43 person detections) for the detection stage. The tracking stage resulted in 10 person tracks, 135 object tracks. The system additionally recorded 275 timeline events and 276 reconstruction steps. These results demonstrate that EvidenceSync can generate structured outputs for the examined CCTV analysis workflow, although these recorded counts cannot yet establish the accuracy of the underlying components.The quantitative evaluation provides further insight into the individual AI components. The YOLO11n object detection evaluation recorded a precision of 58.8%, a recall of 44.1%, an mAP@50 of 46.8%, and an mAP@50-95 of 32.6%. The results show the aspects that need to be improved to increase the object detection performance. The ResNet50-based person re-identification model recorded Rank-1 accuracy of 84.12% and mAP of 66.53%; these results confirm that the model should be used for generating candidate person associations, but these candidates should be validated before they can be interpreted as evidence of identity.The audio-event detection component recorded a Macro-F1 score of 27.21%, a Micro-F1 score of 43.81%, and mAP of 45.20%. These results indicate that the audio-event recognition task will require further investigation and

improvement. The synchronization evaluation showed zero error when tested under controlled conditions. However during robustness testing the mean absolute error was 0.240 seconds with a maximum error reaching 0.900 seconds. These results show that evaluating alignment under different conditions is important.The main contribution of EvidenceSync is organizing multiple analysis outputs into an evidence-review workflow. It provides a timeline multi-camera replay, reconstructed movement and a What May Have Happened interface. These features help investigators review observations and build an understanding of a possible incident. The system aims to keep track of how each interpretation connects to supporting evidence and to make uncertainty and missing information clear.There are still some limitations to consider. The benchmark metrics focus on parts of the system instead of testing the full incident reconstruction workflow. The four-camera session counts do not confirm detection or tracking accuracy. Also the results do not verify all -camera associations or reconstructed event sequences. The 2D movement representation is not based on a calibrated real-world floor plan.Further evaluation should include annotated CCTV recordings, verified event sequences, cross-camera identity matching timestamp accuracy and the quality of the final incident narrative. Testing under recording conditions and with different combinations of evidence will help assess the system's overall reliability.

In summary EvidenceSync shows an approach to analyzing CCTV evidence and reconstructing incidents. Its value lies in helping organize compare and interpret evidence while still requiring verification. The results are a foundation, for improvements not proof that every detection, association or reconstruction is accurate.

7.9 Chapter Summary

This chapter discussed the results of the experimental evaluation of the EvidenceSync CCTV Analysis and Incident Reconstruction System. The evaluation included the analysis of object detection and tracking outputs, camera-wise analysis, unified evidence timeline creation, incident reconstruction, person re-identification, audio-event detection, and camera synchronization.The four-camera session included 20135 frame-level detections. These detections were split between 20092 car detections and 43 person detections. The system tracked 10 person tracks and 135 object tracks. The system produced 275 timeline events and 276 reconstruction steps to enable a chronological review of evidence and incident

analysis.The quantitative evaluation section reported the quantitative results of the evaluation. Specifically, it provided the reported performance of YOLO11n object detection, ResNet50 person re-identification, FSD50K audio-event detection, and camera synchronization.The evaluation results indicate the general usefulness of the unified approach that leverages object detection, tracking, synchronized chronological review of CCTV footage, and incident reconstruction. At the same time, the evaluation results should be considered with the limitations described in this chapter in mind. In particular, while the paper demonstrated the ability of the system to synchronize video, its ability to reconstruct movement is not accurate enough to represent a real-world floor plan. Furthermore, camera-specific tracking does not enable positive identification of persons, and automatic detection does not replace human inspection. A subsequent evaluation using independent annotations will be necessary to assess The accuracy of the whole pipeline.

CHAPTER 8
SUMMARY AND FUTURE WORK

8.1 Summary

This dissertation introduces EvidenceSync, a CCTV analysis and incident reconstruction system. EvidenceSync uses computer vision, object detection, tracking, chronological organization and incident reconstruction to support the review of evidence. Its purpose is to support the review of CCTV footage from multiple cameras to identify activities and examine potential event sequences that are related to the available evidence.The first component of EvidenceSync utilises OpenCV to enable video processing and You Only Look Once (YOLO) 11n and ByteTrack for object detection and tracking. Detection and tracking data can be used to support the analysis of people, cars, and other detectable objects in CCTV footage. The system processes footage from four cameras, including Camera 01, Camera 02, Camera 07, and Camera 10.The second component builds on the tracking and detection data to arrange them in a unified evidence timeline that arranges the data into chronological events and incorporates relevant meta-data such as camera-specific information. This enables a chronological analysis of events and the identification of specific time points that can be used to focus the review of source videosAdditionally component enables timed comparison of data from cameras to spot different evidence. Component also helps build a story of what happened by arranging detection, tracking and event evidence, in the What May Have Happened interface. Component focuses on putting events in order looking at evidence and finding explanations.. Component is made to help examine evidence, not to prove that a crime happened or what someone intended.In particular, the identification of observed, possible, and incriminating evidence must be differentiated.The four-camera session analysed in this dissertation includes 20,135 recorded high-level object detection and tracking events, including 20,092 car detections and 43 person detections. Furthermore, the session includes 10 person tracks, 135 object tracks, 275 timeline events, and 276 reconstruction steps. The numbers reflect the recorded video analysis events and should not be used to infer the total number of actual persons or objects in the environment.EvidenceSync uses FastAPI and Uvicorn to provide an API backend and HTML, CSS and JavaScript to provide a front-end interface. Overall, the system demonstrates the ability to analyse CCTV footage, identify objects, track

them across multiple video streams, organise them into a chronological evidence timeline, and support the reconstruction of possible sequences of events based on available evidence. However, the accuracy of the output is dependent on the source footage, detection and tracking performance, temporal consistency, and available evidence.The system provides a framework for CCTV evidence analysis that enables the chronological organisation of video content from multiple cameras. Using such a framework, investigators can systematically go through all recorded content to identify activities of interest, including those that take place in multiple areas under different viewpoints. It is important to note that the evidence must be interpreted with care. The conclusions must be supported by the review of CCTV footage and other evidence. This helps make sure everything is clear and correct. Looking at the footage and checking proof helps avoid mistakes. We should not jump to conclusions without checking all the facts. Doing this step by step makes the results more reliable.

8.2 Conclusion

The creation of EvidenceSync proves the concept that object detection and tracking, unified CCTV evidence review, timeline construction, and incident reconstruction can be combined in one application for video surveillance. Unlike separate video streams from multiple sources, which are difficult to organize and evaluate, the described tools help structure the obtained data and identify potential connections. The first two parts focus on object detection and tracking. These are used to pull out information, about objects seen in each video. The unified timeline shows this information in the order it happened over time. The synchronized review lets users look at detection and tracking results side by side to compare them. At the end the incident reconstruction tool helps put all the details together in a logical order. This makes it easier to understand what happened during the event. The results of the session illustrate how the app can detect objects, track their movement, display timeline events based on videos. The given outputs confirm the ability of EvidenceSync to build a consistent story out of the collected evidence. However, the number of detected items, tracked objects, and incident steps is not enough to verify the accuracy of results and their correspondence with real-time events.It is necessary to note that each step of the technology has potential issues that should be considered when interpreting results. First, the accuracy of object detection and tracking is highly likely to depend on the quality of scenes, lighting, objects' size, and angles. Moreover, the tracking IDs in different cameras probably have no meaning in determining the correspondence of objects. In turn, the movement of items in space needs to be adjusted to scales to be applied to

a topological map correctly. Finally, the timeline's accuracy depends on the correctness of all preceding operations and proper timestamps. Thus, EvidenceSync is a useful instrument for reviewing CCTV footage and unifying the evidence in one interface. However, it should be used with caution, and more research is needed to confirm the accuracy of results and their relevance to real-world scenarios. The current version of the software is great for structuring the available data, but it has little sense in trying to reconstruct incidents without accurate video evidence.

8.3.1 Limitations

The performance of the system depends on the quality and characteristics of the CCTV footage. Variations in lighting, camera resolution, viewing angle, motion blur and how visible objects are can affect the reliability of object detection and tracking. The four-camera session that was evaluated shows how the system works with that set of videos but it does not prove that the performance of the system is the same in all surveillance environments. Additional testing on videos, from locations, camera types, weather conditions and activity levels is needed to judge how well the performance of the system generalises.

8.3.2 Object Detection Limitations

YOLO11n can provide detection results that're useful for downstream tasks. YOLO11n also supports tracking and analysis. However, it is possible that detections are missed or incorrectly associated when objects are partially hidden, distant, overlapping with other objects, or poorly lit. There may also be false detections that provide irrelevant information to downstream processes.As a result, detection results should be considered as model observations that should be validated against ground-truth annotations and appropriate metrics to ensure accuracy.

8.3.3 Object Tracking Limitations

ByteTrack is applied to link detections across the frames, and to maintain tracks of objects in the video, but tracking errors can occur when objects are temporarily invisible, overlapping each other or moving fast, or leaving or entering the field of view of the camera which can cause a track to be split or merged, and track IDs changed.

Additionally, the track IDs of objects detected by different cameras do not individually identify an object which has been detected by more than one camera, and additional logic is needed to establish these associations and verify them.

8.3.4 Cross-Camera Association Limitation

Although the system allows for the review of footage from multiple cameras, this does not necessarily confirm the association of events depicted in the footage with different cameras. Variations in the angle, lighting, resolution, and overall visibility of the camera may make it difficult to determine the identity of a person or vehicle depicted in the footage.Therefore, if there is no verified mechanism of cross-camera association, the system should not treat the juxtaposition of two observations as confirmation of their association. The association of two observations must always be supported by sufficient visual, chronological, and contextual evidence and should be reviewed by humans.

8.3.5 Timeline and Timestamp Limitations

The unified evidence timeline sorts the recorded events according to the timestamps and the event details provided by the system. It, therefore, relies on the accuracy of the source timestamps, video metadata, frame sampling process, and synchronization settings.It is significant to consider such factors as clock differences in cameras, metadata discrepancies, skipped frames, or an incorrect conversion of timestamps into the timeline. Even though testing synchronization can yield positive results, it is not sufficient to define the operational CCTV session's events in real-time correctly.It is essential to perform independent timestamp checks and comparisons to ensure that all the recorded details are correctly placed in the timeline.

8.3.6 Incident Reconstruction Limitations

The incident reconstruction component arranges the available observations into a particular order. However, the available footage may miss some relevant action or viewpoints. It can be partially obscured, outside cameras' field of view, or insufficient for determining a particular context.The movement recreated in the video is not necessarily a calibrated real-world floorplan if cameras were not calibrated and mapped. Likewise, a particular suggested course of events is not necessarily indicative of the actual cause, criminal intent, or responsibility. The

reconstruction should always be based on the available evidence, and the ambiguous information should be reviewed by a human investigator.

8.3.7 Evaluation and Generalization Limitations

The reported detection, tracking, timeline, and reconstruction counts represent outputs of the evaluated session. They cannot be used to establish detection accuracy, tracking accuracy, identity association accuracy, or the fact of correct incident reconstructed.Component level benchmarks are also not indicative of end-to-end performance on the target CCTV session. A model's accuracy on a benchmark dataset does not imply that it will perform similarly on the videos in the project.A more complete evaluation requires manually annotated reference data, rigorous evaluation procedures, and measurements of individual components and end-to-end performance. Until such a process is in place the system should be considered an investigative aide rather than a fully autonomous incident verification system.

8.3.8 Operational and Human Verification Limitations

EvidenceSync is intended to aid evidence examination, but the interpretation of CCTV footage is the responsibility of the investigator. Model-generated detections, track associations, event sequences, and reconstructed narratives can contain errors or omit important context. Its outputs should thus be cross-referenced in relation to the original video before being used to support investigative conclusions. Practical deployment would also require evidence integrity, access control, auditability, privacy, storage, preservation procedures around the source footage, and the procedures used to preserve and review it.

8.4 Future Work

Although EvidenceSync provides an integrated workflow for CCTV analysis and incident reconstruction, many improvements can be explored to improve its accuracy, robustness, scalability, and practical usefulness. Future work will try to confirm the abilities Future work will solve the problems mentioned in this chapter and Future work will grow the system so that each conclusion can still be traced back, to the original evidence.

8.4.1 Improved Object Detection Performance

Future work can explore how well YOLO11n works in surveillance situations such as bad lighting, motion blur, partial blocking, crowds and faraway shots. We could test YOLO11n on an more varied set of CCTV videos to see when it misses detections or gets them wrong. We could improve YOLO11n by fine-tuning it on CCTV footage using good image preprocessing and comparing it to other model setups. Every change, to YOLO11n must be tested on an annotated set of videos measuring precision, recall and mean Average Precision (mAP). This will show whether the changes give measurable gains instead of just looking better by eye.

8.4.2 Robust Tracking and Cross-Camera Identity Association

Future development could make track maintenance better when objects are temporarily hidden or leave the camera view and return later. Different tracking setups and appearance-based features could be tested to reduce tracks and incorrect connections. Another area to explore is a -camera matching system for people and vehicles. This system would compare appearance timing overlaps possible camera transitions and other available data to find likely matches. These matches should include confidence levels or uncertainty values. They should be easy for an investigator to review. Evaluation must rely on labeled identities across cameras and clear defined metrics. Track IDs from cameras should not be combined just because they look similar or happen at the same time. Improvements, like this would help determine if a person or vehicle can be traced across camera views.

8.4.3 Real-World Spatial Calibration and Movement Reconstruction

The current movement reconstruction can be improved to show movement in a calibrated space. In the future we could add camera calibration fix perspective issues use reference points and include a mapped floor plan. This would give us the camera settings and measurements. With calibration we can turn image coordinates into a shared spatial reference. This lets us compare movements across cameras in a meaningful way. We can also show which areas each camera covers and how views change from one, to another. This helps us know where each observation was recorded.The final spatial model would give a picture of how movement happens across the whole area. reconstruction must be checked against known reference positions and movements.Until that calibration is carried out. Verified, reconstructed movement should keep being shown as a visual or relative view and should not be treated as

real-world coordinates.

8.4.4 More Accurate Evidence Timeline and Synchronization

I think future work could improve the reliability of the evidence timeline. We could validate camera timestamps, spot inconsistencies. Find synchronization errors. The unified evidence timeline could also compare camera timestamps with reference events or with timing information that was checked by people when that information is available. I also think more tests could measure how much synchronization error exists across video lengths, frame rates, sample settings and clock offsets. An event-level check could show whether the unified evidence timeline keeps the order of events that were marked by different people. Finally the unified evidence timeline could have filters for camera object tracks, event type and time range. This would help us find evidence while keeping a link, between each timeline entry and its source video segment.

8.4.5 Evidence-Grounded Incident Reconstruction

The incident reconstruction component could be improved by making sure each proposed event sequence is clearly connected to the evidence that supports it. Every step in the reconstruction should include details like the source camera, the timestamp or time interval the relevant video frame or segment and any track information. It should also note if there is uncertainty in the data. Future work could separate events that are directly observed from those that are inferred based on relationships, between events. When the evidence is incomplete or conflicting the system should highlight the uncertainty of presenting one single sequence as final. Investigators should have tools to correct or reject any reconstruction steps that are proposed. To check how effective these improvements are the system-generated reconstructions should be compared with reference timelines made independently and with event sequences reviewed by experts. The goal is to improve traceability, consistency and reviewability. This does not mean automated incident reconstruction will replace judgment.

8.4.6 Multimodal Evidence Integration

An additional research direction is looking at combining types of evidence when there is good

data and proven ways to process it. Examples are sounds that happen in an environment, written notes, from speech, records of who accessed something and other time-based information that's useful in an investigation. This combining could help people looking into something check what they see against recorded facts. For example a sound that happens at the time as a video clip could be shown together if the times are accurate.. Just because two things happen close together does not mean they are connected. Future work needs to look at ways to match types of evidence deal with unsure timing handle when information contradicts itself and show the evidence in a clear way. Every type of evidence should be checked on its own before looking at how it helps in the process.

8.4.7 Comprehensive Evaluation and Benchmarking

An additional research direction is looking at combining types of evidence when there is good data and proven ways to process it. Examples are sounds that happen in an environment, written notes, from speech, records of who accessed something and other time-based information that's useful in an investigation. This combining could help people looking into something check what they see against recorded facts. For example a sound that happens at the time as a video clip could be shown together if the times are accurate.. Just because two things happen close together does not mean they are connected. Future work needs to look at ways to match types of evidence deal with unsure timing handle when information contradicts itself and show the evidence in a clear way. Every type of evidence should be checked on its own before looking at how it helps in the process.

8.4.8 Scalability and Performance Optimization

The system could be made to handle videos and more camera feeds while still letting users review evidence quickly. In the future we could look into video decoding, smarter frame sampling combining detections in batches improving how tracking works how data is stored and using async processing. These steps would help find where the system slows down.We should also test the system by checking how long it takes to process how memory it uses how much storage it needs and how fast it responds. As videos get longer and more cameras are added. This would show us how power and space the system needs and what’s the most it can handle.The backend could be updated to run processing jobs reliably show clear progress updates recover from errors and manage many analysis tasks, at once. These changes would

make the workflow better for investigations while keeping all evidence and results consistent..

8.4.9 Evidence Integrity, Security, and Auditability

Since the system is handling surveillance footage that might be important for investigations future development should focus on improving how evidence is managed and how access is controlled. Possible changes could include adding hashes to the original files and to any evidence that is exported. The system could also keep track of actions taken by users limit access during investigations and make sure that any results are stored securely. The system could keep a record of when evidence was added, which settings were used for processing and how the reconstructed outputs were made or checked. If an investigator makes changes, to the results the system could keep the version and also note the changes that were made. These features would help people make sure that the original files and any results have not been changed without anyone knowing. These changes would need to be tested as part of a review of how evidence is handled and how secure the system is.

8.4.10 Human-in-the-Loop Investigation Workflow

Future work can make it easier for investigators to look at and check the results from AI. A review interface can let users say if a detection is correct or not fix the time when something happened add notes about movement paths and change the steps used to rebuild events. The system can focus on events that're not clear or might be important but it should not use a score from the model as the chance that a crime took place. Investigators should be able to watch the video and see other details before agreeing with an idea. Testing can look at whether these review tools make checking evidence more consistent and faster. The main goal is to use computer work along with checks, by people making sure that the final decisions are based on real proof.

CHAPTER 9
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APPENDIX A

SOURCE CODE IMPLEMENTATION OF EVIDENCESYNC

A.1 CCTV Video Processing and Frame Sampling

Future work can make it easier for investigators to look at and check the results from AI. A review interface can let users say if a detection is correct or not fix the time when something happened add notes about movement paths and change the steps used to rebuild events. The system can focus on events that're not clear or might be important but it should not use a score from the model as the chance that a crime took place. Investigators should be able to watch the video and see other details before agreeing with an idea. Testing can look at whether these review tools make checking evidence more consistent and faster. The main goal is to use computer work along with checks, by people making sure that the final decisions are based on real proof.

Source file: backend/app/api/reconstruction_session_api.py

```
def analyze_video(
    model: YOLO,
    path: Path,
    camera_id: str,
) -> dict[str, Any]:

    capture = cv2.VideoCapture(str(path))

    if not capture.isOpened():
        raise RuntimeError(
            f"Unable to open CCTV video: {path.name}"
        )

    fps = float(
        capture.get(cv2.CAP_PROP_FPS) or 30.0
    )
```

```
    frame_count = int(
        capture.get(cv2.CAP_PROP_FRAME_COUNT) or 0
    )

    width = int(
        capture.get(cv2.CAP_PROP_FRAME_WIDTH) or 640
    )

    height = int(
        capture.get(cv2.CAP_PROP_FRAME_HEIGHT) or 360
    )

    duration = (
        frame_count / fps
        if fps > 0 and frame_count > 0
        else 0.0
    )

    capture.release()

    stride = max(
        1,
        int(round(fps / ANALYSIS_FPS))
    )
```

The procedure obtains the videos properties. The procedure calculates the sampling stride.For example, when the source video runs at 30 frames per second and the configured analysis rate is 2 frames per second, the calculated stride is 15 frames.

A.2 Object Detection and Tracking

EvidenceSync uses the YOLO model and the ByteTrack tracking configuration to process relevant object classes in CCTV footage. The implementation also carries out a separate person detection pass. This helps keep person detections that might not appear in the multi-class tracking output.

```
results = model.track(
    source=str(path),
    tracker="bytetrack.yaml",
    classes=list(RELEVANT_CLASSES.keys()),
    conf=CONFIDENCE_THRESHOLD,
    iou=IOU_THRESHOLD,
    persist=True,
    stream=True,
    verbose=False,
    vid_stride=stride,
)

person_model = model.__class__(str(MODEL_PATH))

person_results = person_model.predict(
    source=str(path),
    classes=[0],
    conf=0.10,
    iou=IOU_THRESHOLD,
    stream=True,
    verbose=False,
    vid_stride=stride,
)
```

The primary tracking pass processes the configured object classes and generates tracking results. The additional prediction pass focuses on the person class. The resulting detections are combined during subsequent processing. When a person detection does not have a native tracker ID, the implementation can attempt a tentative association with a recent detection from the same camera. Such associations are not treated as verified identities.

A.3 Evidence Timeline Generation

The build_timeline() function converts the observed person and object tracks into structured events. It records when a person or object was first observed and, where applicable, when it was last observed. Each event contains fields such as timestamp, camera ID, event type, state,

and track ID.

Source file: backend/app/api/reconstruction_session_api.py

```
def build_timeline(
    camera_results: list[dict[str, Any]],
) -> list[dict[str, Any]]:

    events = []

    for camera in camera_results:
        camera_id = camera["camera_id"]

        for person in camera["observed_people"]:
            points = person.get("points", [])

            if not points:
                continue

            first_time = float(points[0][0])
            last_time = float(points[-1][0])

            events.append({
                "timestamp": first_time,
                "camera_id": camera_id,
                "event_type": "PERSON_APPEARED",
                "state": "OBSERVED",
                "track_id": person["track_id"],
                "description": (
                    f"Person track "
                    f"{person['track_id']} first observed."
                ),
            })

            events.append({
                "timestamp": last_time,
```

```
            "camera_id": camera_id,
            "event_type": "PERSON_LAST_OBSERVED",
            "state": "OBSERVED",
            "track_id": person["track_id"],
            "description": (
f"Person track "
                f"{person['track_id']} last observed."
            ),
        })

    return events
```

Note: This is an abridged illustration of the function. The actual implementation also processes object tracks and activity-review candidates. Use the complete function from the source file if your dissertation requires a full code listing.

The timeline provides a common event structure for reviewing observations from different cameras. Events marked OBSERVED represent recorded observations, whereas inferred activity candidates are represented separately with an INFERRED state.

A.4 Incident Reconstruction

The build_reconstruction_steps() function converts timeline events into readable reconstruction steps. It creates descriptions for person appearances, last observations, object detections, and activity-review events. If no reconstructable events are available, the function adds a corresponding unavailable-event entry.

```
def build_reconstruction_steps(
    timeline: list[dict[str, Any]],
) -> list[dict[str, Any]]:

    steps = []

    for event in timeline:
        event_type = event["event_type"]

        if event_type == "PERSON_APPEARED":
```

```
        text = (
            f"Person track {event['track_id']} "
            f"was observed by {event['camera_id']}."
        )

    elif event_type == "PERSON_LAST_OBSERVED":
        text = (
            f"Person track {event['track_id']} "
            f"was last observed at this point "
            f"in the available camera coverage."
        )

    elif event_type == "OBJECT_APPEARED":
        text = (
            f"YOLO11n detected a "
            f"{event['object_class']} "
            f"on camera-local track "
            f"{event['track_id']}."
        )

    else:
        text = event.get(
            "description",
            "Evidence event."
        )

    steps.append({
        "timestamp": event["timestamp"],
        "camera_id": event["camera_id"],
        "event_type": event["event_type"],
        "state": event["state"],
        "track_id": event.get("track_id"),
        "object_class": event.get("object_class"),
        "text": text,
```

```
        })

    return steps
```

The generated steps present detected events in a readable format while preserving their timestamps and camera references. The implementation also includes a human-review notice stating that the available evidence does not independently establish identity, intent, ownership, causation, guilt, or criminal responsibility.

A.5 FastAPI Backend Integration

EvidenceSync uses FastAPI to provide backend endpoints and register the application's API routers. During application startup, the database initialization function is called. A health-check endpoint is also provided to return the service status.

Source file: backend/app/main.py

```
app = FastAPI(
    title="EvidenceSync",
    version="0.1.0",
    description="Multimodal Evidence Investigation Platform",
)

@app.on_event("startup")
def startup_event():
    init_db()

app.include_router(evidence_router)
app.include_router(case_summary_router)
app.include_router(timeline_router)
app.include_router(graph_router)
app.include_router(investigation_router)
app.include_router(model_router)
app.include_router(unified_timeline_router)
app.include_router(incident_profile_router)
app.include_router(hypothesis_router)
app.include_router(reconstruction_router)
```

```
app.include_router(tracking_router)
app.include_router(reconstruction_session_router)
app.include_router(forensic_intelligence_router)

@app.get("/health")
def health_check():
    return {
        "status": "OK",
        "service": "EvidenceSync",
        "version": "0.1.0",
    }
```

The backend integrates the evidence, timeline, tracking, and reconstruction components through registered API routers. Database initialization supports the application's data layer, while the health endpoint provides a basic service-status response.

A.6 Summary

This appendix presents selected implementation excerpts from EvidenceSync. The code demonstrates CCTV video-property extraction, frame sampling, YOLO-based detection and tracking, timeline event generation, reconstruction-step preparation, and FastAPI backend integration. These components support the organization and review of CCTV evidence. Their outputs should be interpreted as computer-vision observations and investigative aids, with final conclusions subject to human verification.

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